
EEM 561 Machine Vision

Lecture 4

Color

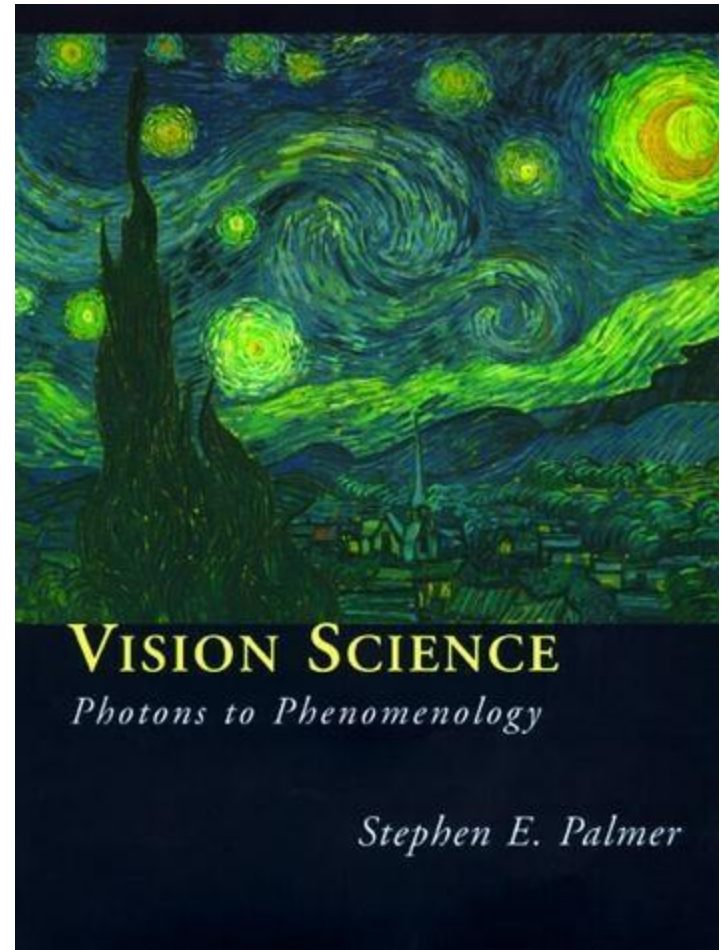
Spring 2025

Asst. Prof. Dr. Onur Kılınç

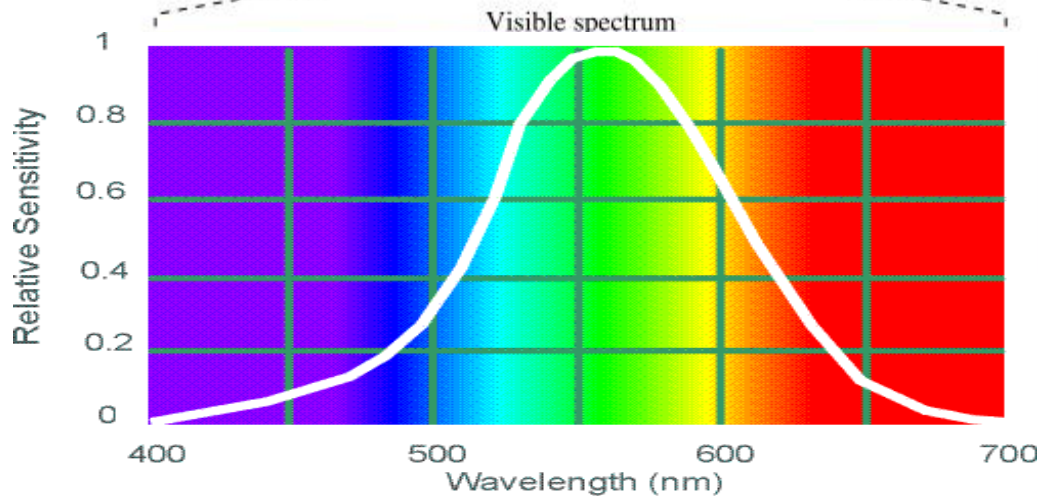
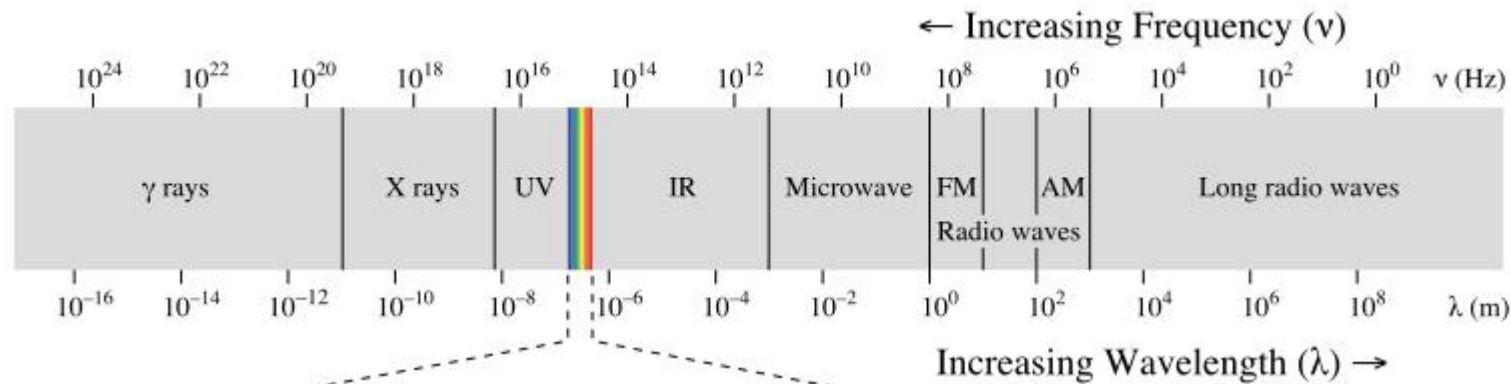
Eskisehir Tech. Uni., Dept. EEE

What is color?

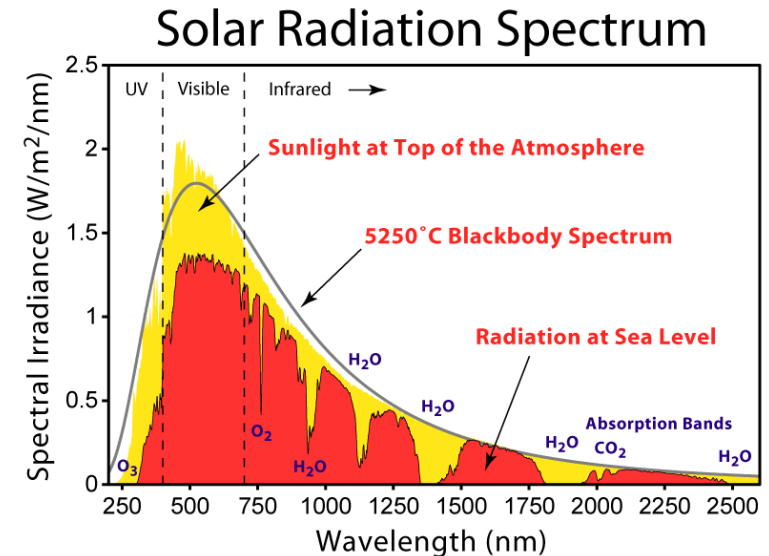
- Color is the result of interaction between physical light in the environment and our visual system
- Color is a psychological property of our visual experiences when we look at objects and lights, *not* a physical property of those objects or lights (S. Palmer, *Vision Science: Photons to Phenomenology*)



Electromagnetic spectrum

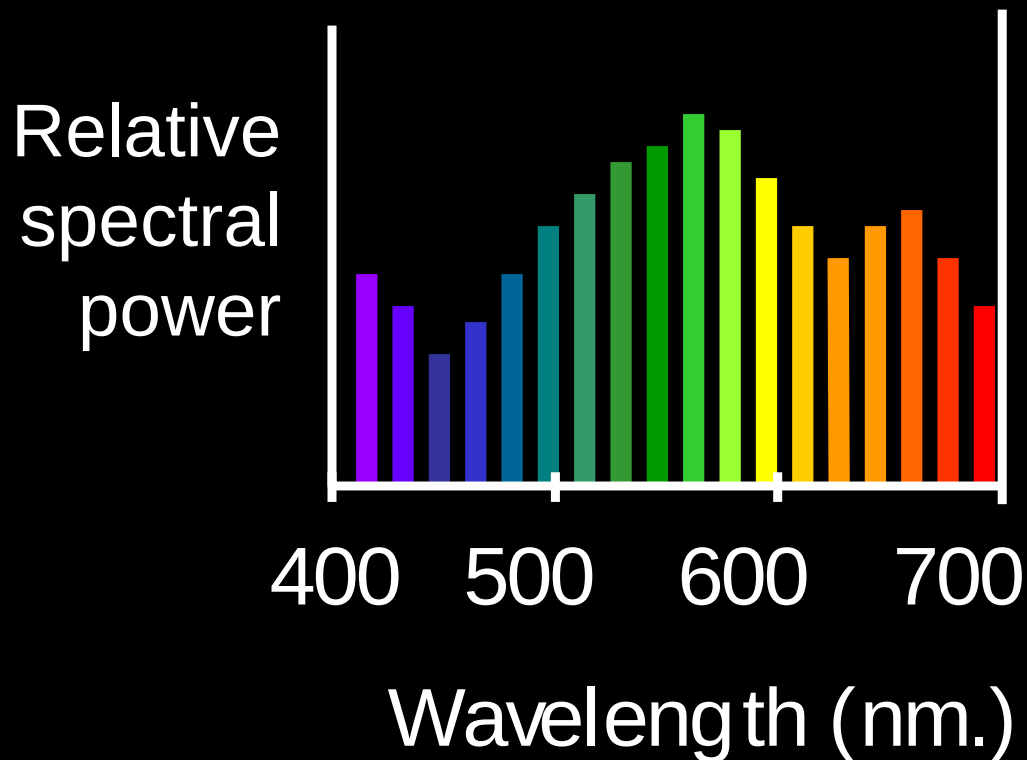


Human Luminance Sensitivity Function



The Physics of Light

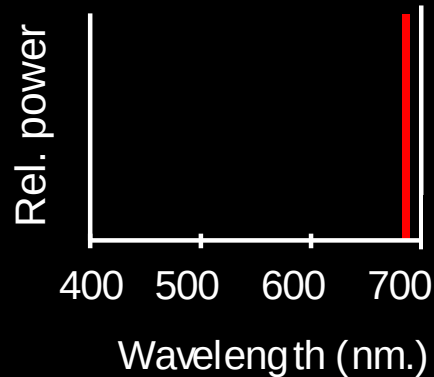
Any source of light can be completely described physically by its spectrum: the amount of energy emitted (per time unit) at each wavelength 400 - 700 nm.



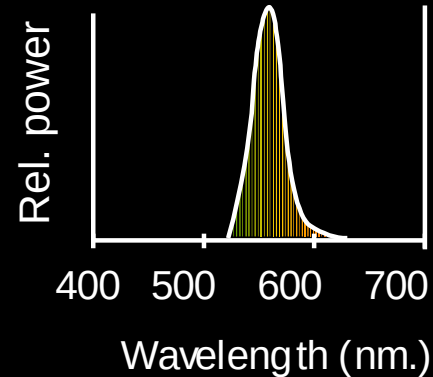
Spectra of Light Sources

Some examples of the spectra of light sources

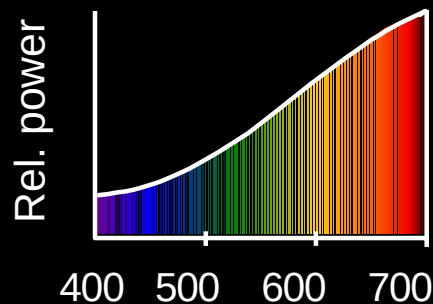
A. Ruby Laser



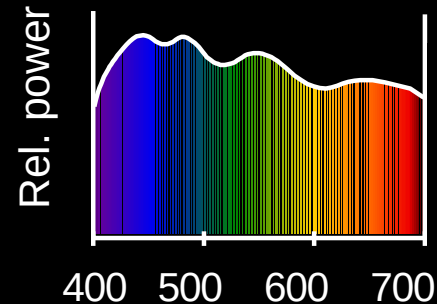
B. Gallium Phosphide Crystal



C. Tungsten Lightbulb

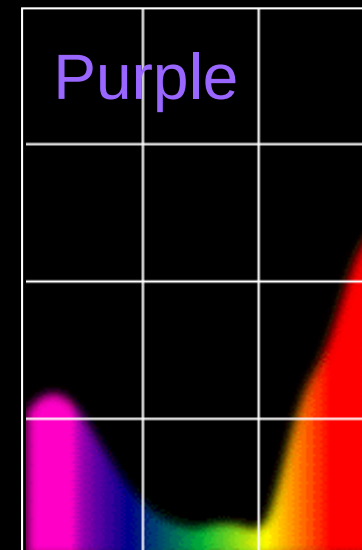
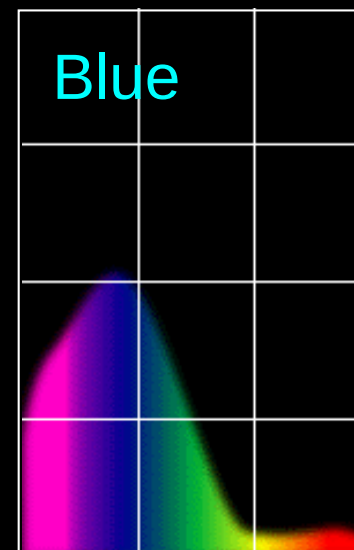
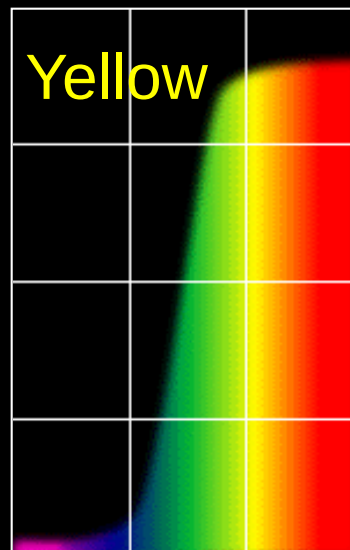
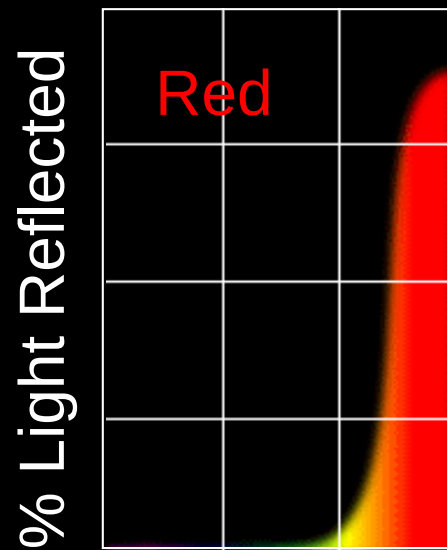


D. Normal Daylight



Reflectance Spectra of Surfaces

Some examples of the reflectance spectra of surfaces



400

700

400

700

400

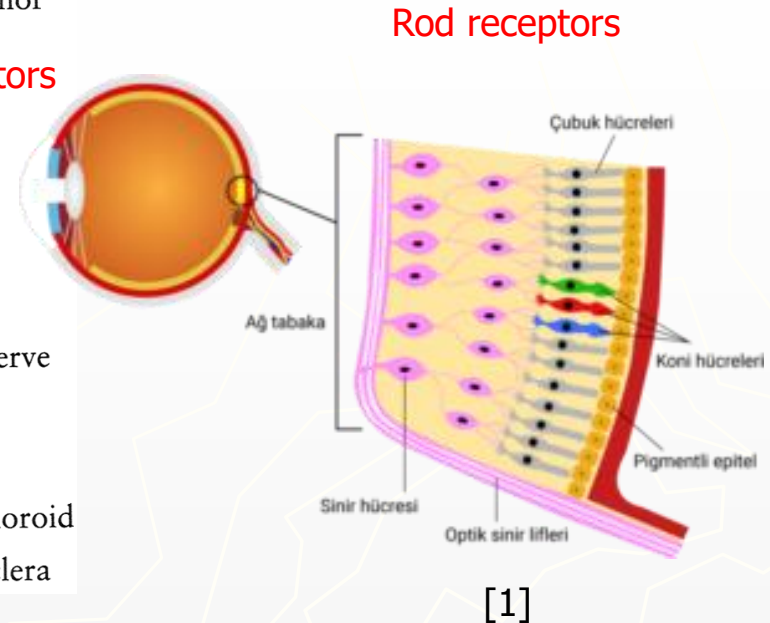
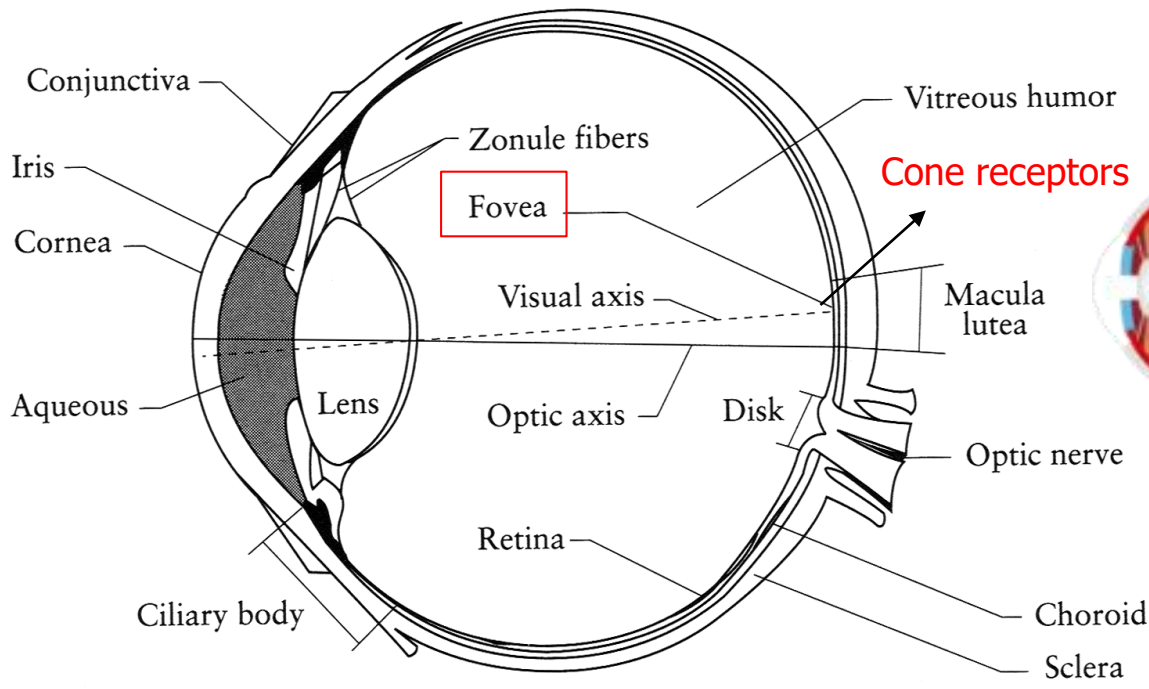
700

400

700

Wavelength (nm)

The Eye



The human eye is a camera!

- **Lens** - changes shape by using ciliary muscles (to focus on objects at different distances)
- **Pupil** - the hole (aperture) whose size is controlled by the iris
- **Iris** - colored annulus with radial muscles
- **Retina** - photoreceptor cells

Some Typical Ranges of illumination

► Illumination

Lumen — A unit of light flow or luminous flux

Lumen per square meter (lm/m^2) — The metric unit of measure for illuminance of a surface

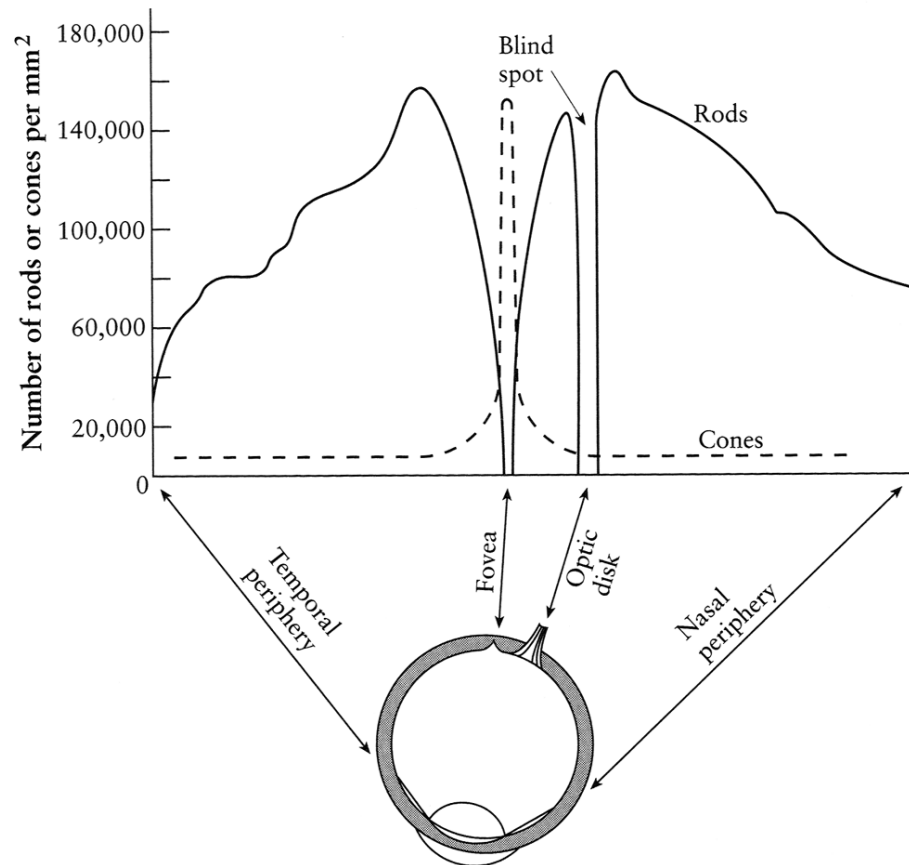
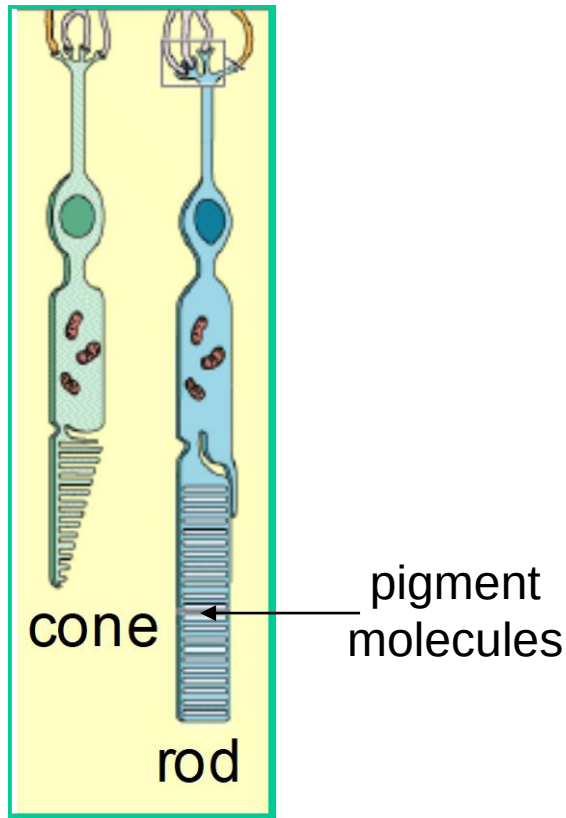
- On a clear day, the sun may produce in excess of 90,000 lm/m^2 of illumination on the surface of the Earth
- On a cloudy day, the sun may produce less than 10,000 lm/m^2 of illumination on the surface of the Earth
- On a clear evening, the moon yields about 0.1 lm/m^2 of illumination
- The typical illumination level in a commercial office is about 1000 lm/m^2

Some Typical Ranges of Reflectance

► Reflectance

- 0.01 for black velvet
- 0.65 for stainless steel
- 0.80 for flat-white wall paint
- 0.90 for silver-plated metal
- 0.93 for snow

Rods and cones, fovea



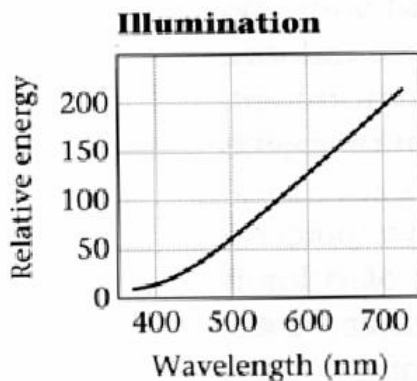
Rods are responsible for intensity, cones for color perception
Rods and cones are *non-uniformly* distributed on the retina

- **Fovea** - Small region (1 or 2°) at the center of the visual field containing the highest density of cones – *and no rods*

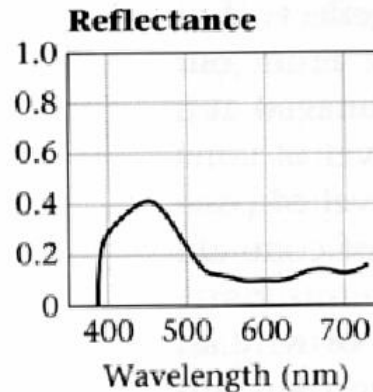
Interaction of light and surfaces



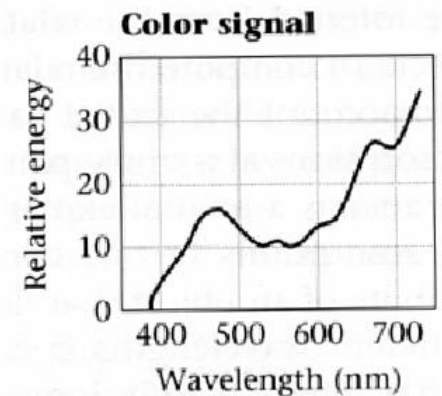
- Reflected color is the result of interaction of light source spectrum with surface reflectance



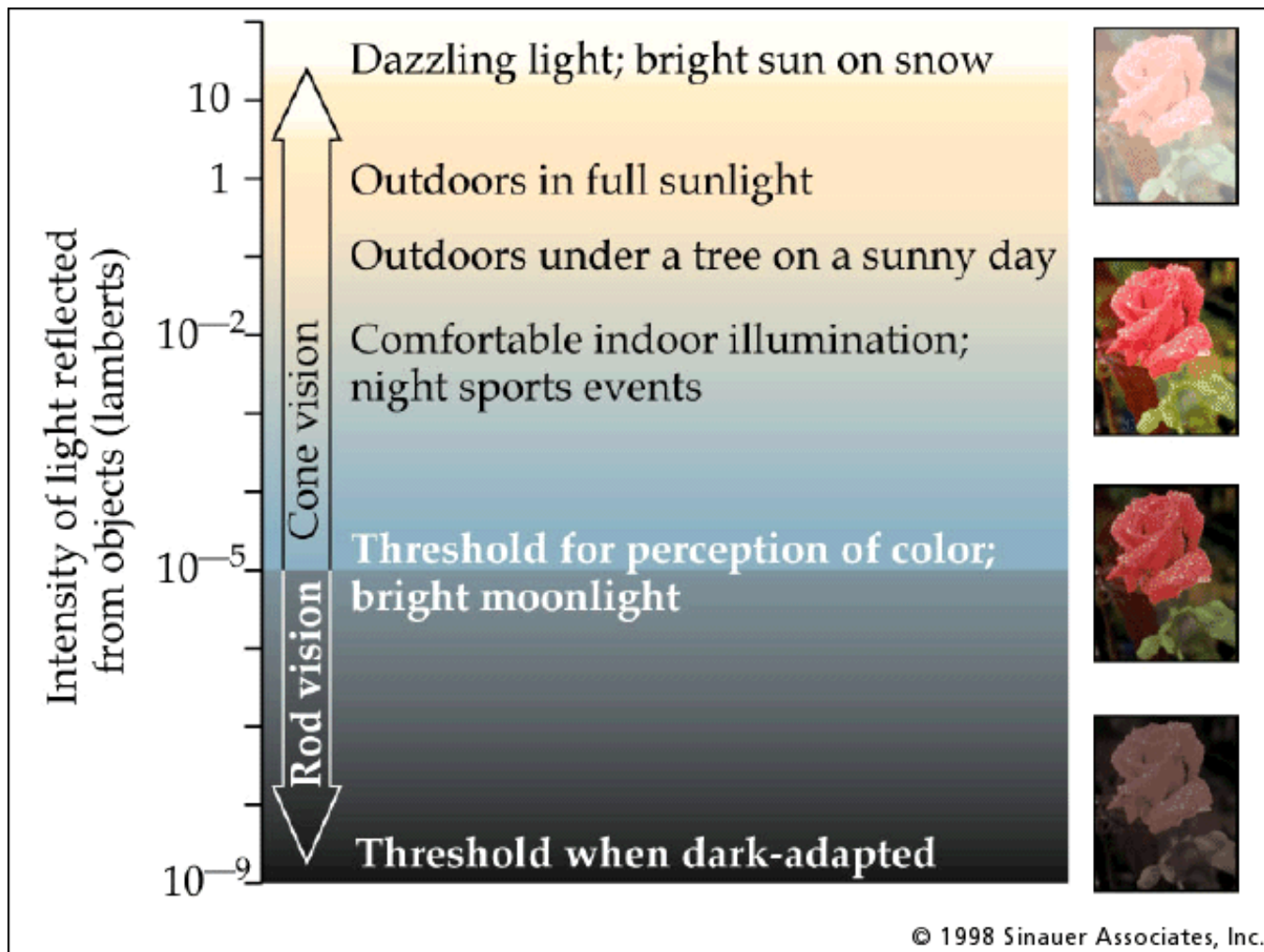
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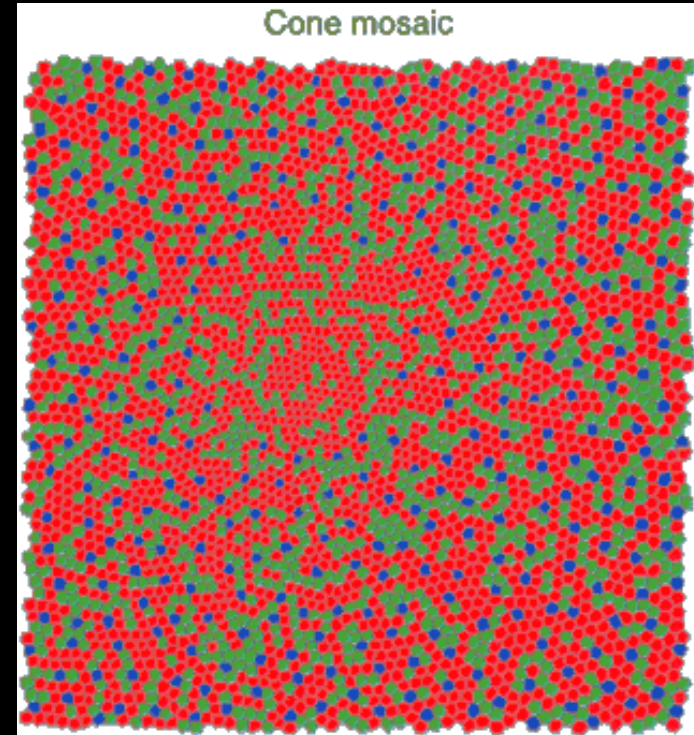
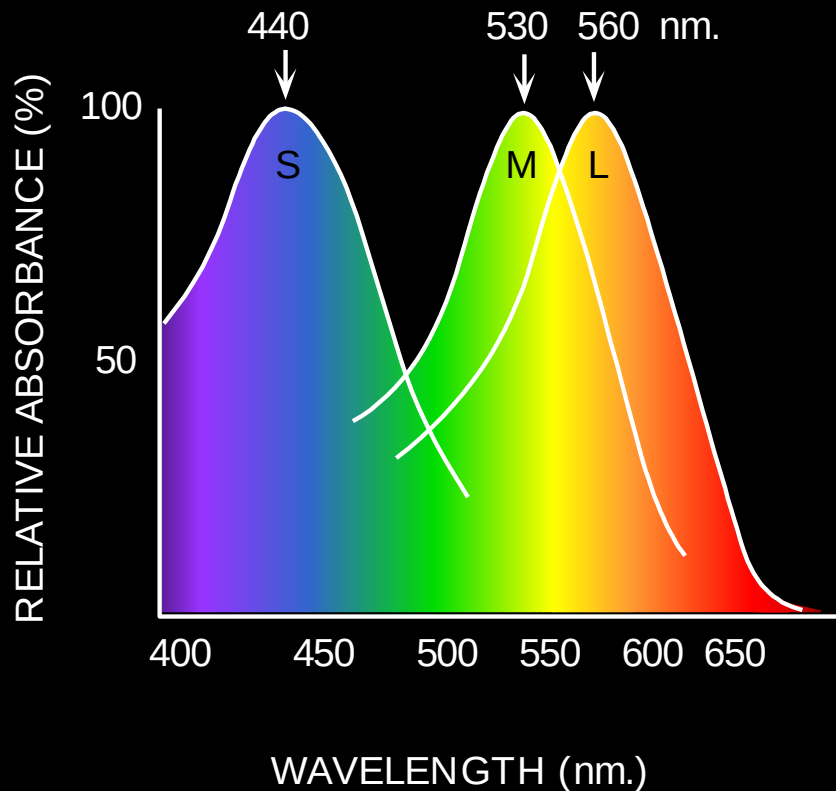
Rod / Cone sensitivity



Why can't we read in the dark?

Physiology of Color Vision

Three kinds of cones:



- Ratio of L to M to S cones: approx. 10:5:1
- Almost no S cones in the center of the fovea

Physiology of Color Vision: Fun facts

► Name the color:



?



?



?

Physiology of Color Vision: Fun facts

► Name the color:



Fuşya



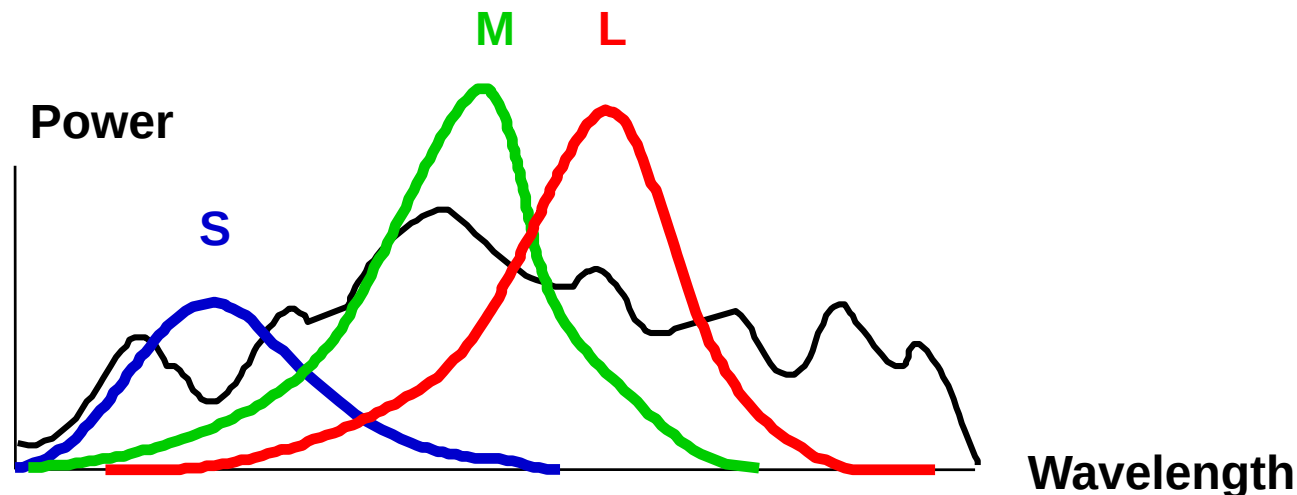
Nar Çiçeği



Vişne Çürüğü

-
- There are S-cone, M-cones, and L-cones for (short, medium, long wavelenghts) in fovea.
 - “M” and “L” pigments are encoded on the X-chromosome
 - That’s why men are more likely to be **color blind**
 - “L” gene has high variation, so some women may be ***tetrachromatic***
 - Some animals have **one** (night **animals**), **two** (e.g., dogs), **four** (fish, birds), **five** (pigeons, some reptiles/amphibians), or even 12 (mantis shrimp) types of cones

Color perception



Rods and cones act as *filters* on the spectrum

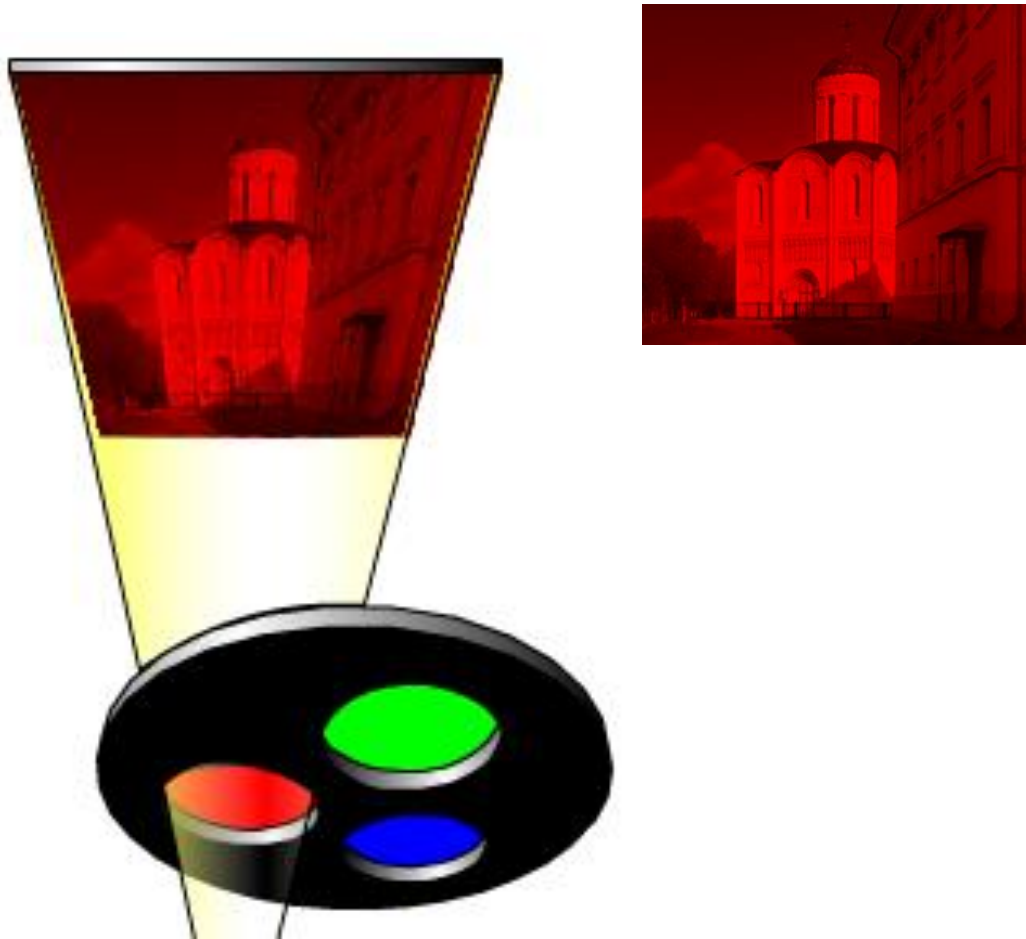
- To get the output of a filter, multiply its response curve by the spectrum, integrate over all wavelengths
 - Each cone yields one number
- How can we represent an entire spectrum with 3 numbers?
- We can't! Most of the information is lost
 - As a result, two different spectra may appear indistinguishable
 - » such spectra are known as **metamers**

Color

So far, we've only talked about monochrome sensors. Color imaging has been implemented in a number of ways:

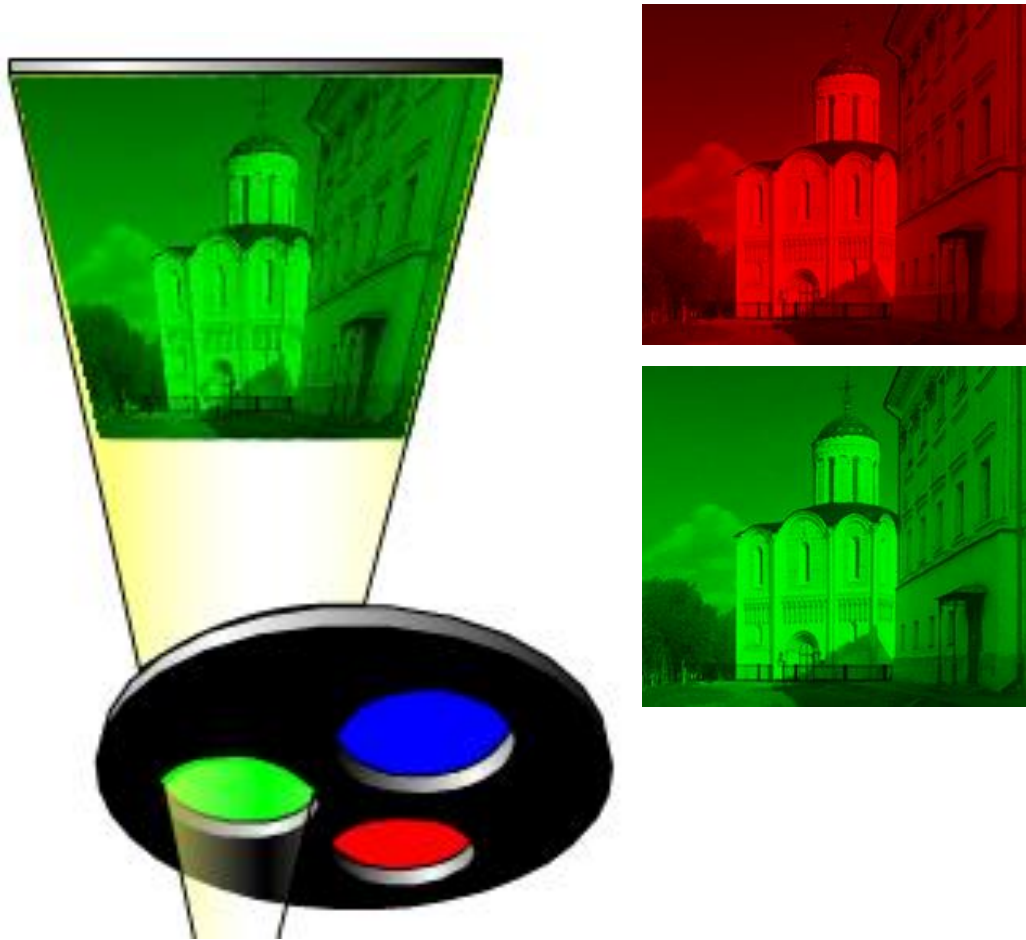
- Field sequential
- Multi-chip
- Color filter array
- X3 sensor

Field sequential



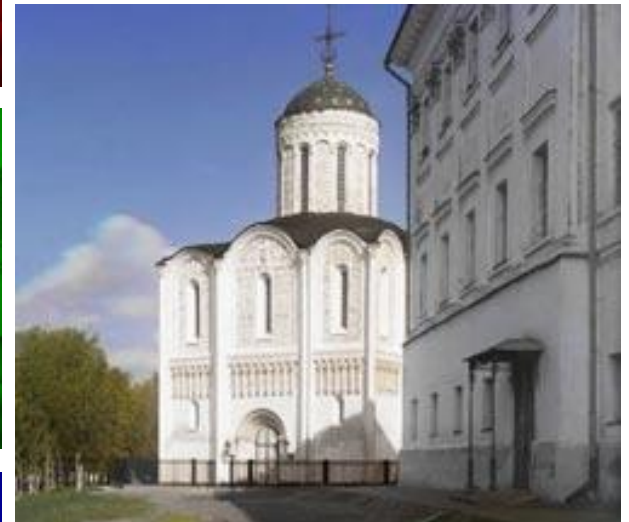
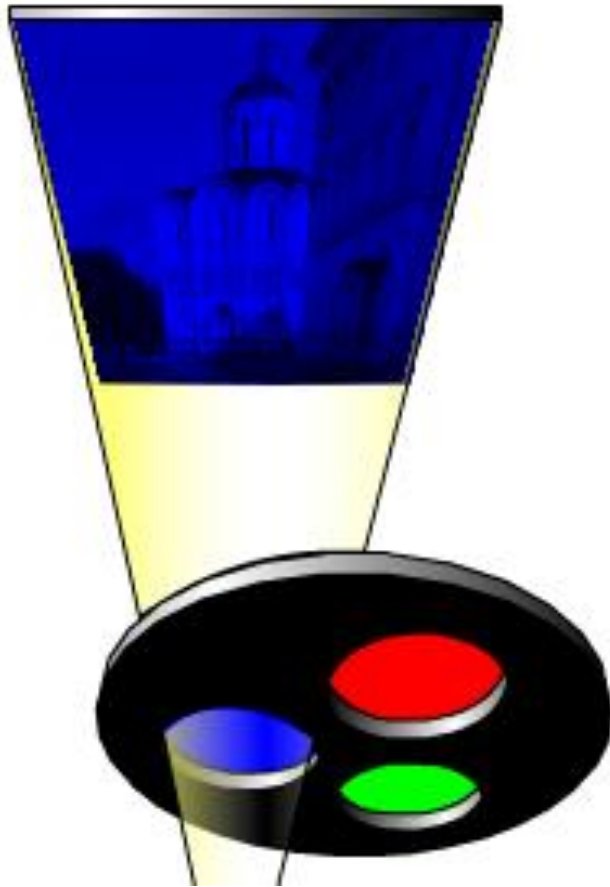
From Prof. Alan C. Bovik

Field sequential

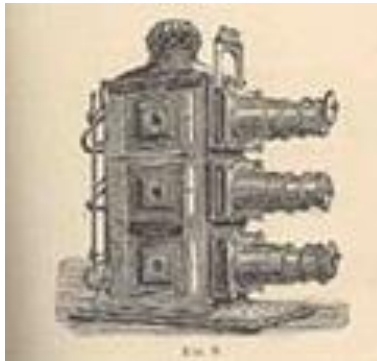


From Prof. Alan C. Bovik

Field sequential



Prokudin-Gorskii (early 1900's)



Lantern
projector

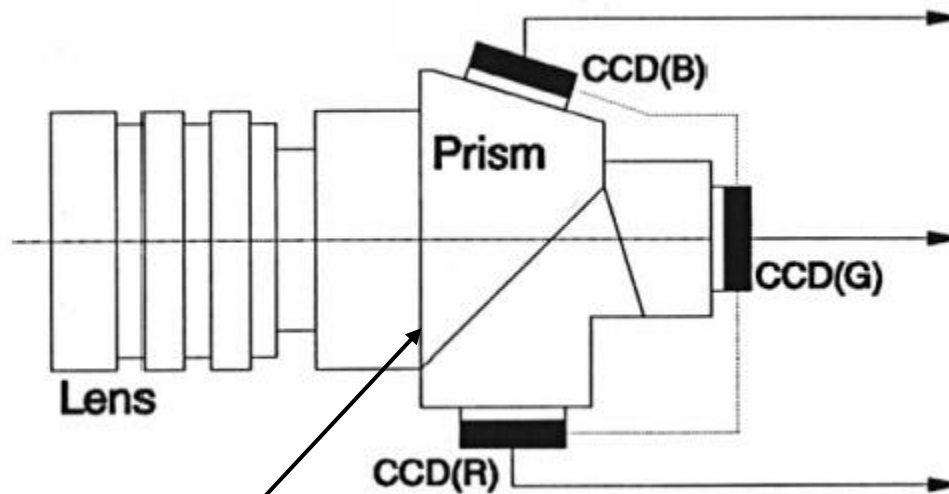


<http://www.loc.gov/exhibits/empire/>

Prokudin-Gorskii (early 1990's)



Multi-chip



wavelength
dependent

Color filter array

R	G	B
R	G	B
R	G	B
R	G	B

R	G	B	G
R	G	B	G
R	G	B	G
R	G	B	G

Stripes

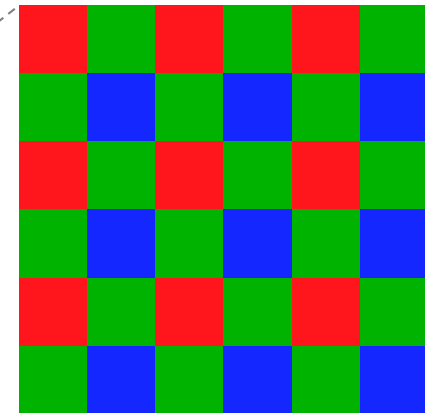
Ye	G	Cy	G
Ye	G	Cy	G
Ye	G	Cy	G
Ye	G	Cy	G

Cy	W	Ye	G
Ye	G	Cy	W
Cy	W	Ye	G
Ye	G	Cy	W

G	Mg	G	Mg
Cy	Ye	Cy	Ye
Mg	G	Mg	G
Cy	Ye	Cy	Ye

Mosaics

R	G	R	G
G	B	G	B
R	G	R	G
G	B	G	B



Bayer pattern

Color filter arrays (CFAs)/color filter mosaics

Color filter array

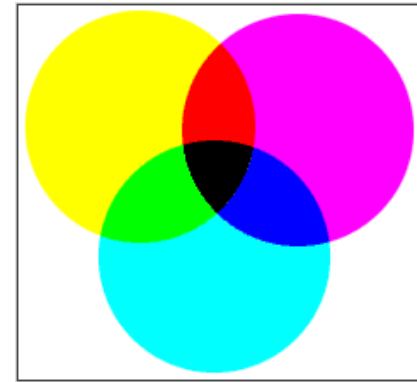
Kodak DCS620x

R	G	B
R	G	B
R	G	B
R	G	B

R	G	B	G
R	G	B	G
R	G	B	G
R	G	B	G

Stripes

Ye	G	Cy	G
Ye	G	Cy	G
Ye	G	Cy	G
Ye	G	Cy	G

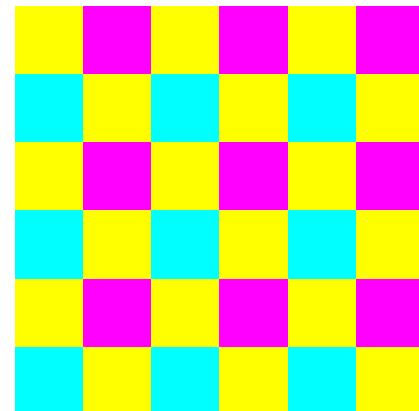


Cy	W	Ye	G
Ye	G	Cy	W
Cy	W	Ye	G
Ye	G	Cy	W

G	Mg	G	Mg
Cy	Ye	Cy	Ye
Mg	G	Mg	G
Cy	Ye	Cy	Ye

Mosaics

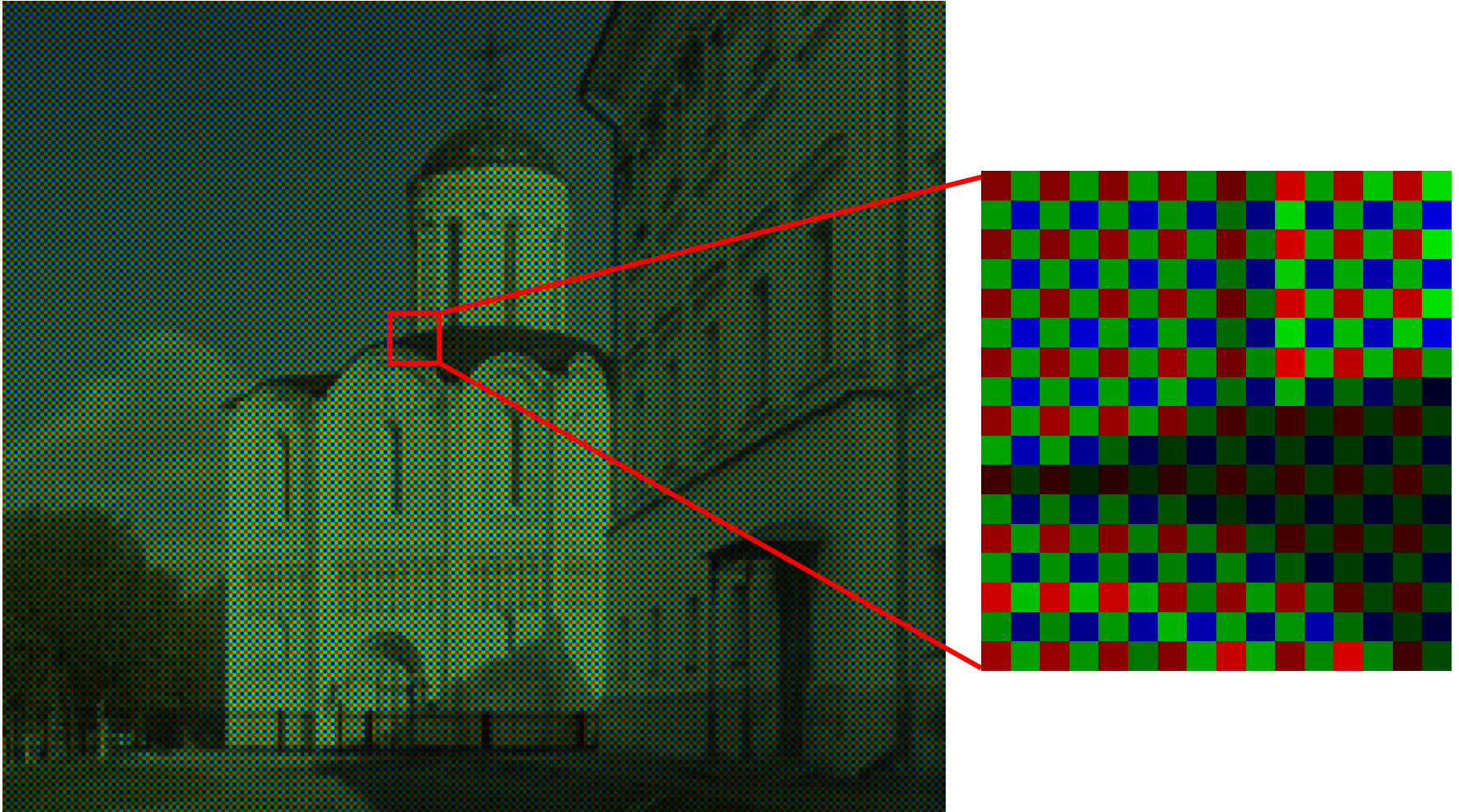
R	G	R	G
G	B	G	B
R	G	R	G
G	B	G	B



CMY

Color filter arrays (CFAs)/color filter mosaics

Bayer's pattern



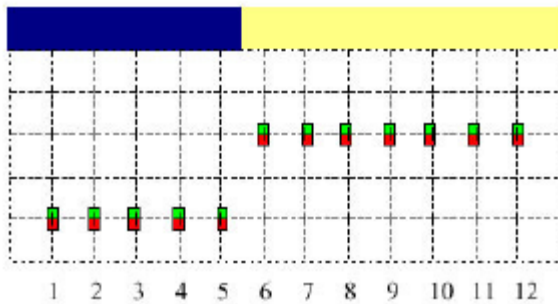
Demosaicking CFA's

R_{11}	G_{12}	R_{13}	G_{14}	R_{15}	G_{16}	R_{17}
G_{21}	B_{22}	G_{23}	B_{24}	G_{25}	B_{26}	G_{27}
R_{31}	G_{32}	R_{33}	G_{34}	R_{35}	G_{36}	R_{37}
G_{41}	B_{42}	G_{43}	B_{44}	G_{45}	B_{46}	G_{47}
R_{51}	G_{52}	R_{53}	G_{54}	R_{55}	G_{56}	R_{57}

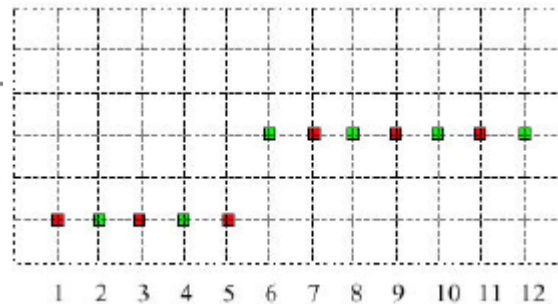
bilinear interpolation

$$G_{44} = (G_{34} + G_{43} + G_{45} + G_{54}) / 4$$

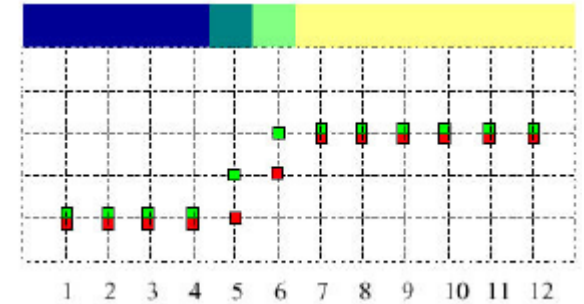
$$R_{44} = (R_{33} + R_{35} + R_{53} + R_{55}) / 4$$



original



input



linear interpolation

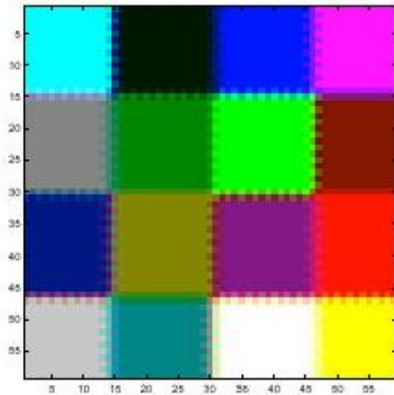
Demosaicking CFA's

R ₁₁	G ₁₂	R ₁₃	G ₁₄	R ₁₅	G ₁₆	R ₁₇
G ₂₁	B ₂₂	G ₂₃	B ₂₄	G ₂₅	B ₂₆	G ₂₇
R ₃₁	G ₃₂	R ₃₃	G ₃₄	R ₃₅	G ₃₆	R ₃₇
G ₄₁	B ₄₂	G ₄₃	B ₄₄	G ₄₅	B ₄₆	G ₄₇
R ₅₁	G ₅₂	R ₅₃	G ₅₄	R ₅₅	G ₅₆	R ₅₇
G ₆₁	B ₆₂	G ₆₃	B ₆₄	G ₆₅	B ₆₆	G ₆₇
R ₇₁	G ₇₂	R ₇₃	G ₇₄	R ₇₅	G ₇₆	R ₇₇

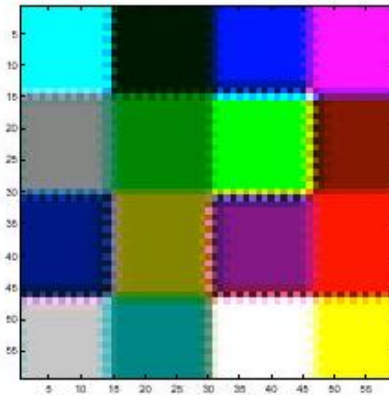
Median-based interpolation
(Freeman)

1. Linear interpolation
2. Median filter on color differences

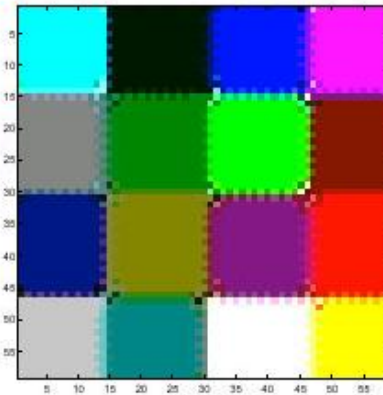
Demosaicking CFA's



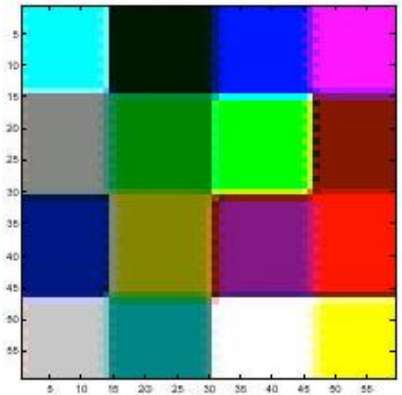
Bilinear



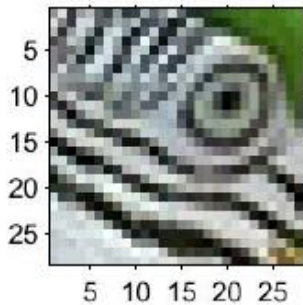
Cok



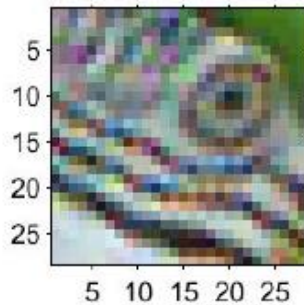
Freeman



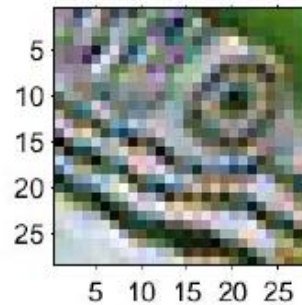
LaRoche



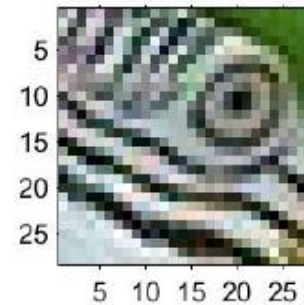
Input



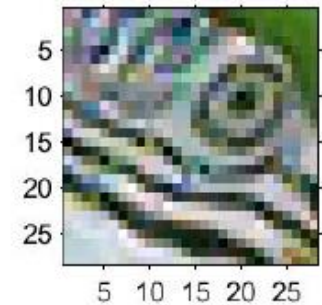
Bilinear



Cok



Freeman



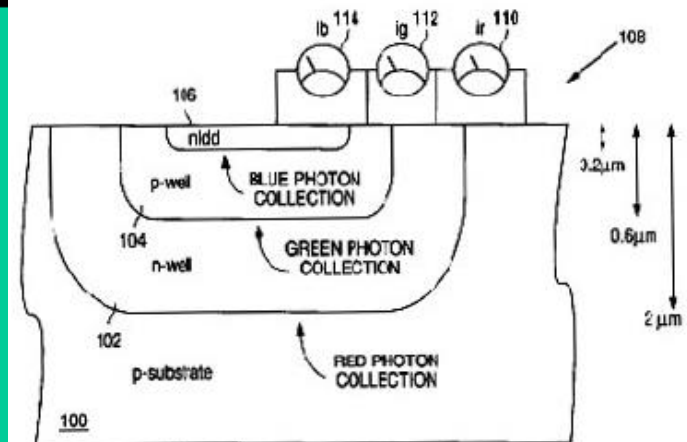
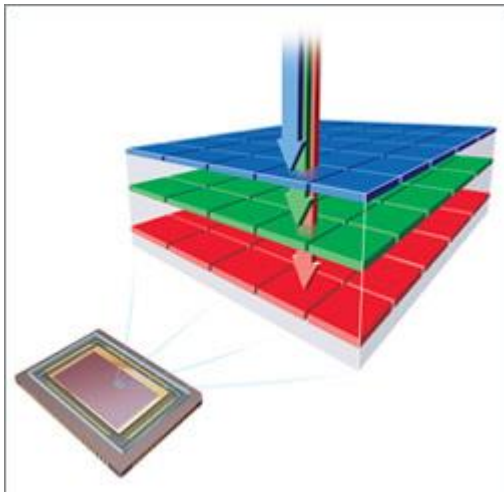
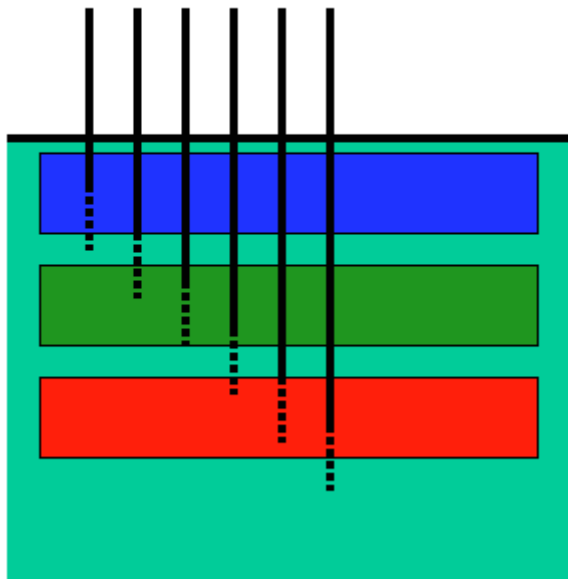
LaRoche

Generally, Freeman's is the best, especially for natural images.

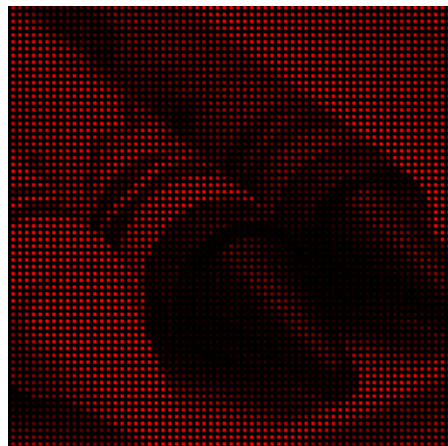
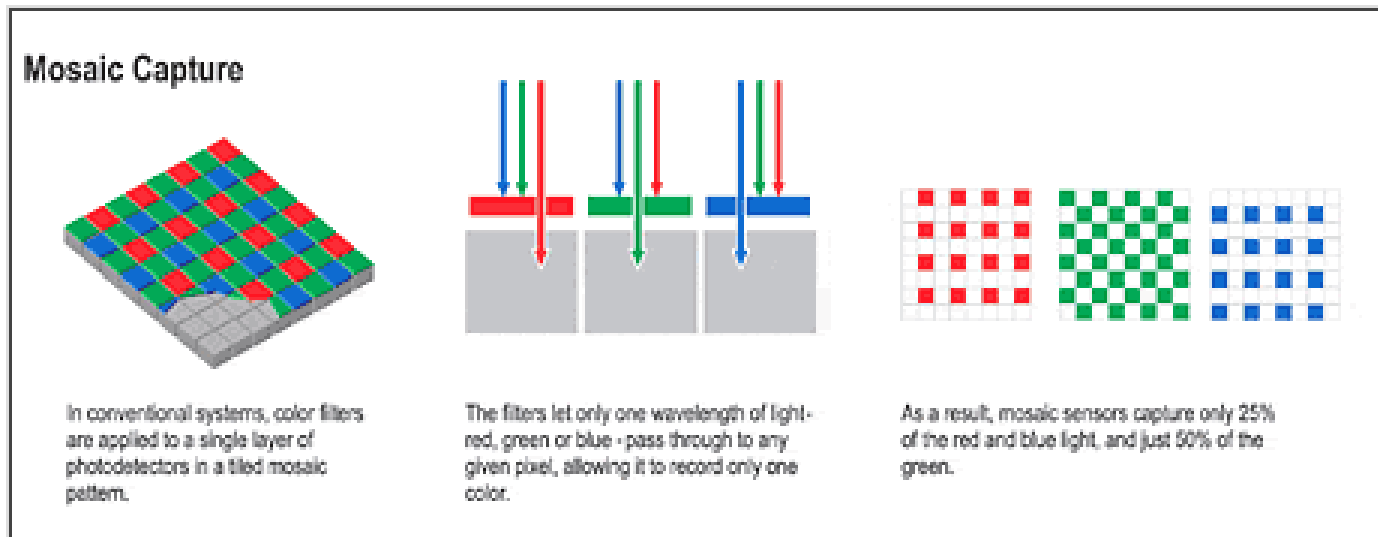
Foveon X3 sensor

- light penetrates to different depths for different wavelengths
- multilayer CMOS sensor gets 3 different spectral sensitivities

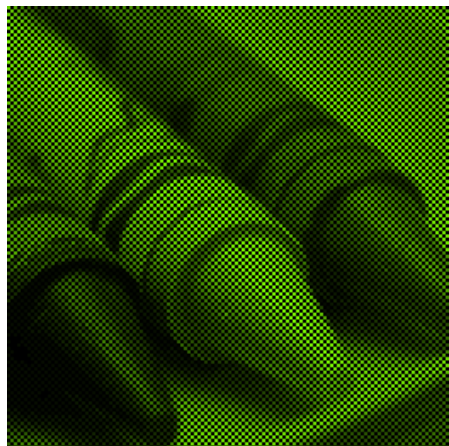
400 700



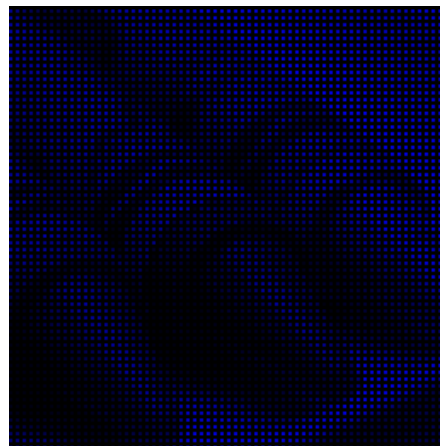
Color filter array



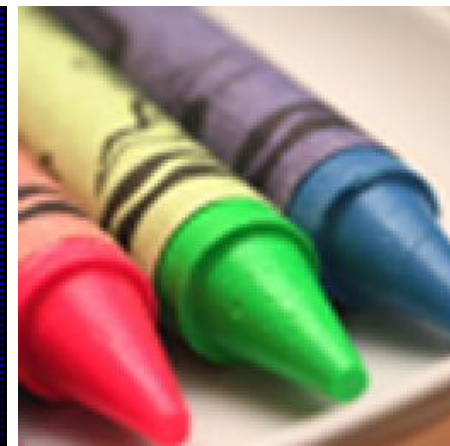
red



green

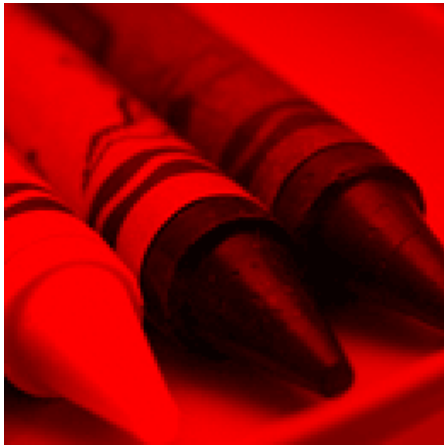
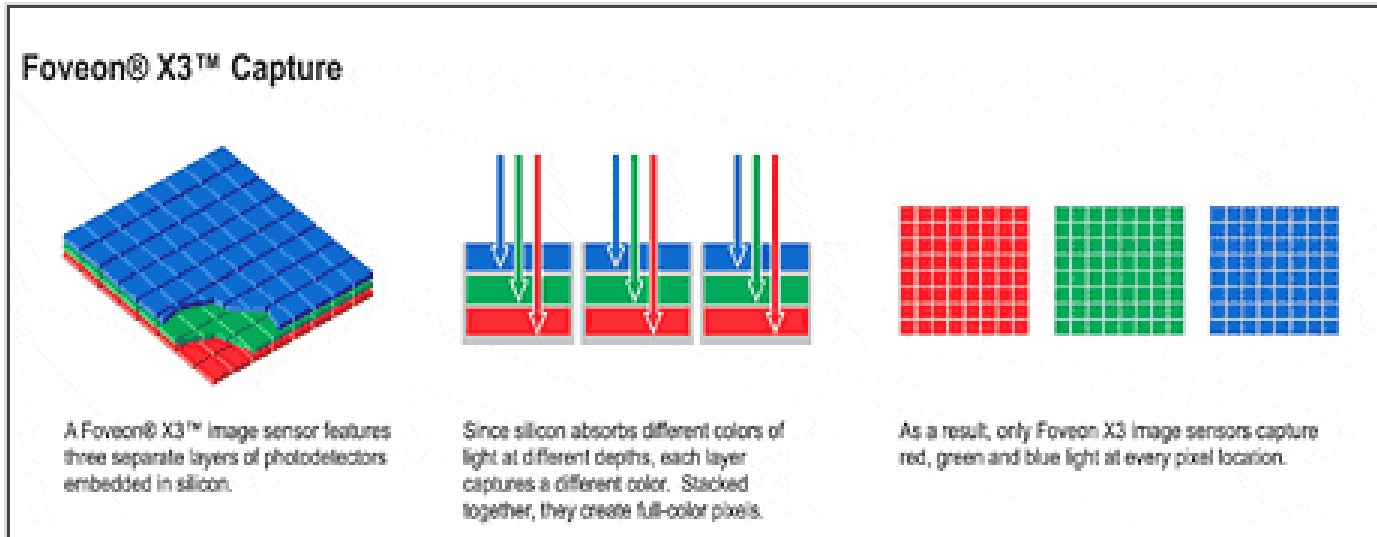


blue



output

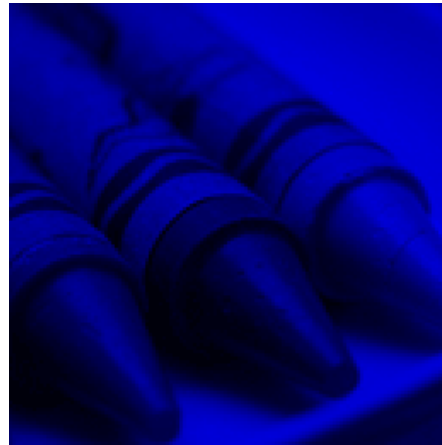
X3 technology



red



green

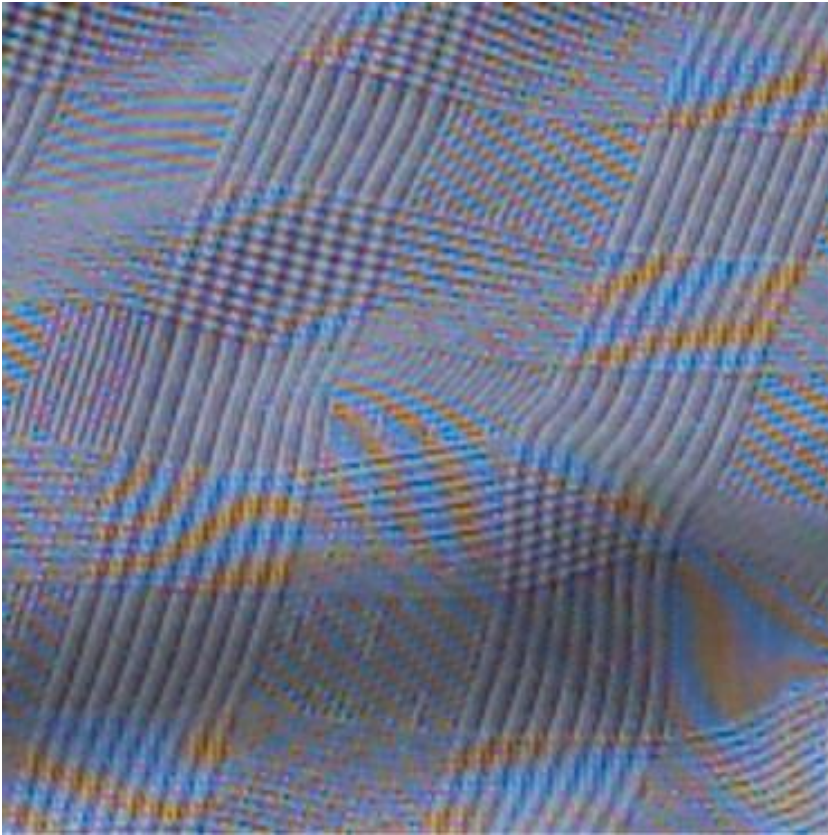


blue

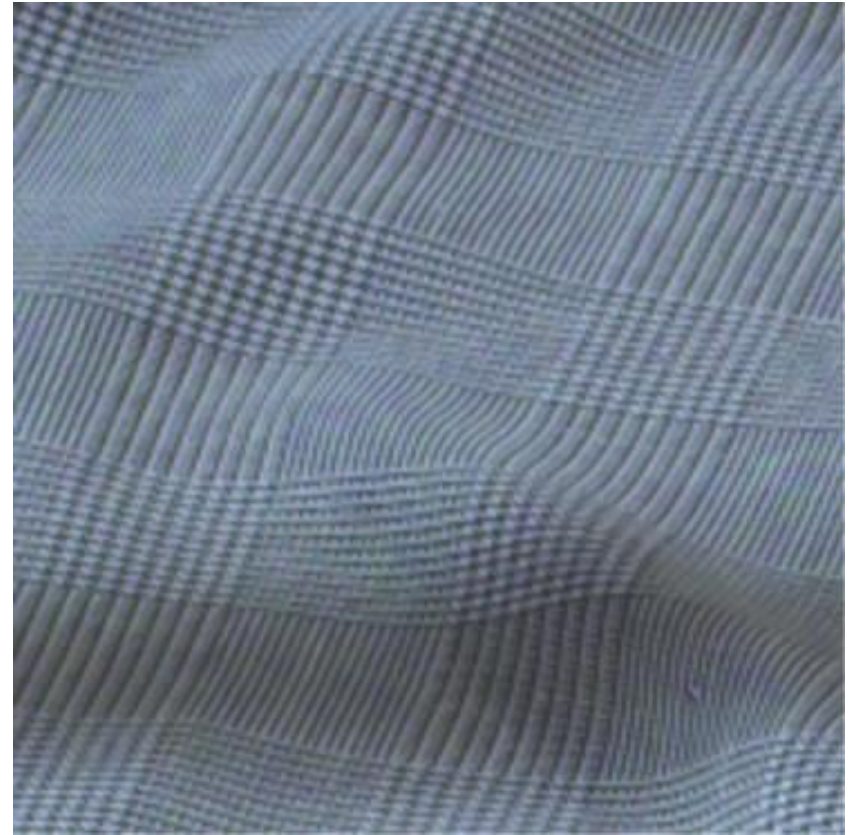


output

Foveon X3 sensor



Bayer CFA



X3 sensor

Cameras with X3



Sigma SD10, SD9



Polaroid X530

Sigma SD9 vs Canon D30



Color processing

- After color values are recorded, more color processing usually happens:
 - White balance
 - Non-linearity to approximate film response or match TV monitor gamma

Auto White Balance



warmer



automatic white balance

$$\begin{bmatrix} R \\ G \\ B \end{bmatrix} = \begin{bmatrix} 255/R'_w & 0 & 0 \\ 0 & 255/G'_w & 0 \\ 0 & 0 & 255/B'_w \end{bmatrix} \begin{bmatrix} R' \\ G' \\ B' \end{bmatrix}$$

Auto White Balance

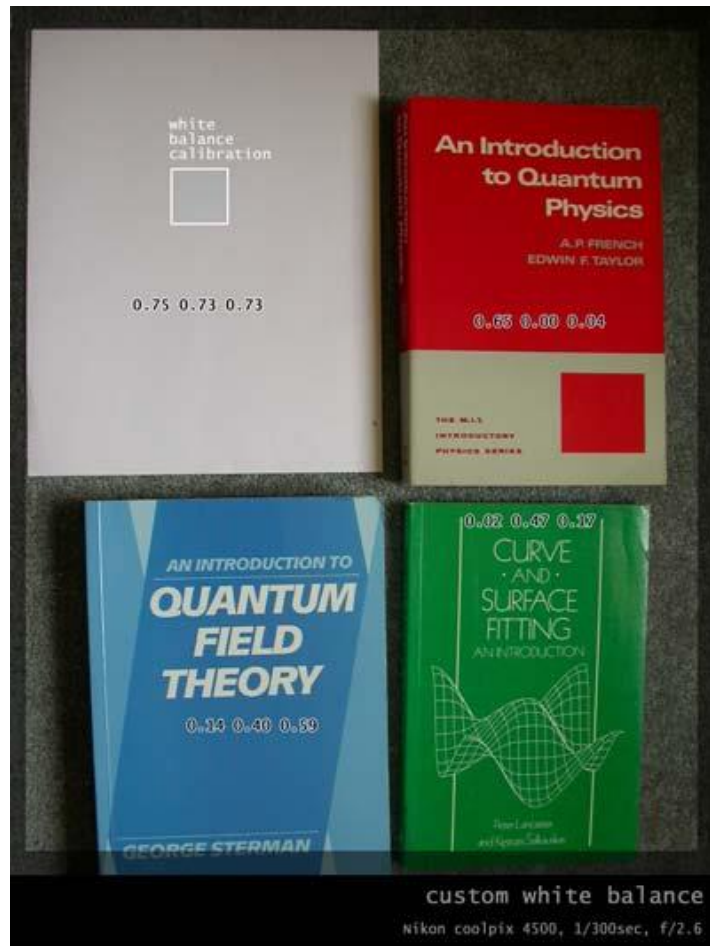


The auto white balance was unable to find a white reference, resulting in dull and artificial colors.



The auto white balance got it right this time in a very similar scene because it could use the clouds as its white reference.

Manual white balance



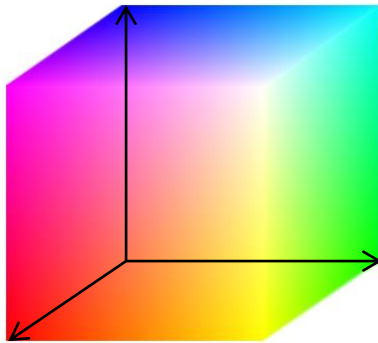
white balance with
the white book



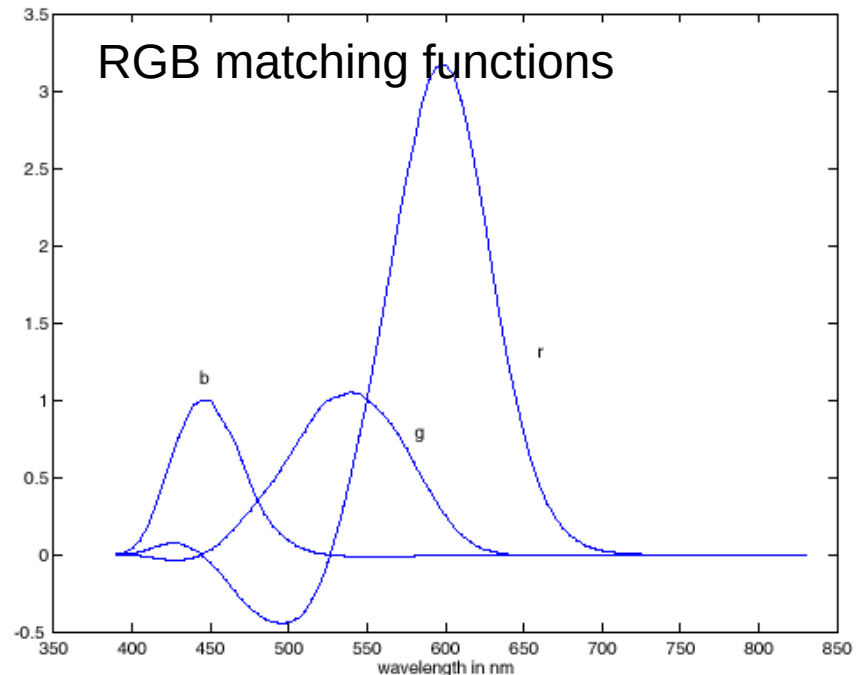
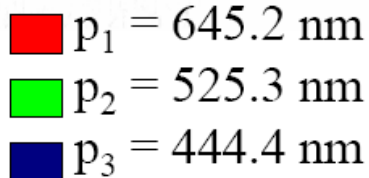
white balance with
the red book

Linear color spaces

- Defined by a choice of three **primaries**
- The **coordinates** of a color are given by the weights of the primaries used to match it
- In addition to primaries, need to specify **matching functions**: the amount of each primary needed to match a monochromatic light source at each wavelength



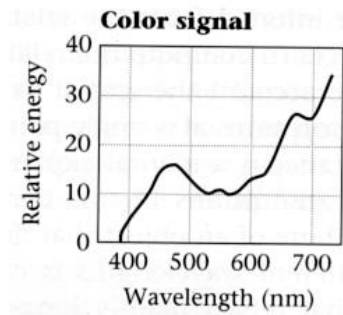
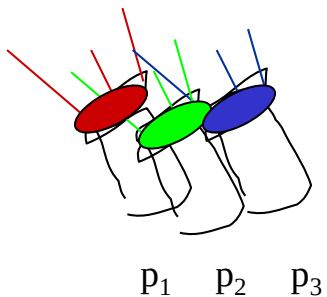
RGB primaries



Linear color spaces

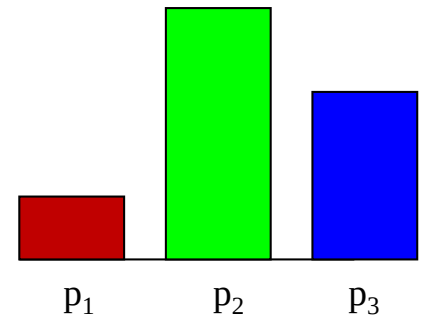
- How to compute the weights of the primaries to match any spectral signal?

Given: a choice of three primaries and a target color signal



?

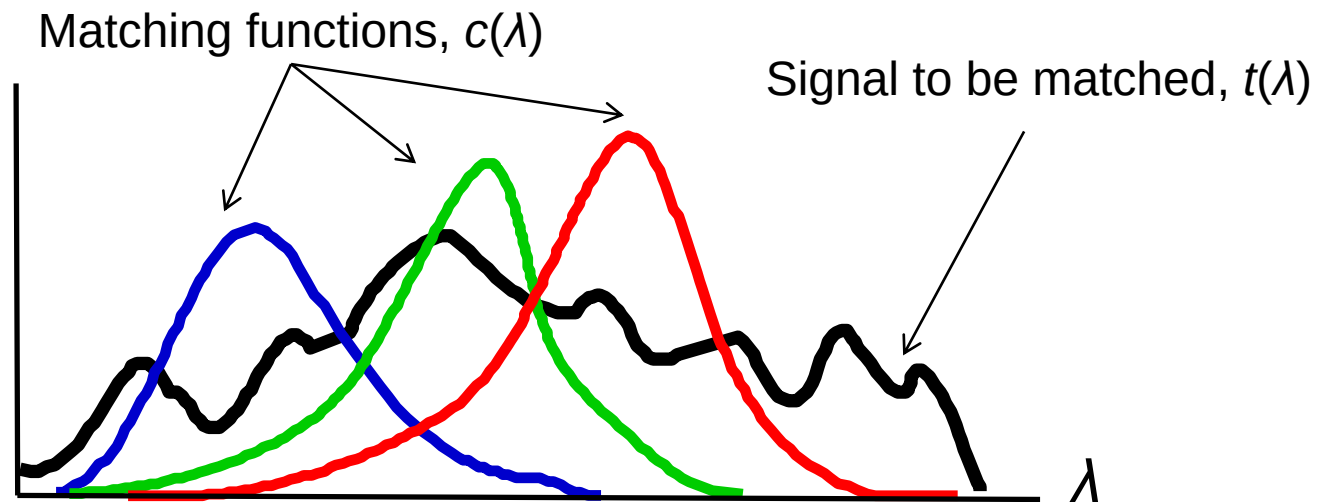
Find: weights of the primaries needed to match the color signal



Linear color spaces

- How to compute the weights of the primaries to match any spectral signal?
- Let $c(\lambda)$ be one of the matching functions, and let $t(\lambda)$ be the spectrum of the signal. Then the weight of the corresponding primary needed to match t is

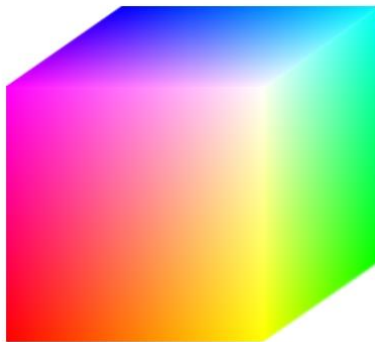
$$w = \int_{\lambda} c(\lambda)t(\lambda)d\lambda$$



RGB space

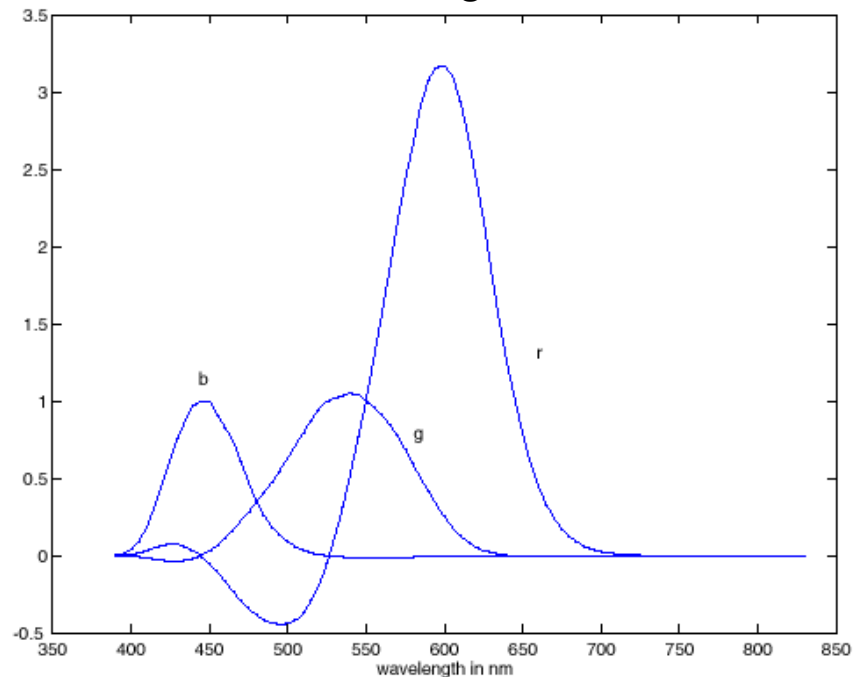
- Primaries are monochromatic lights (for CRT monitors, they correspond to the three types of phosphors)
- *Subtractive matching* required for some wavelengths

RGB primaries



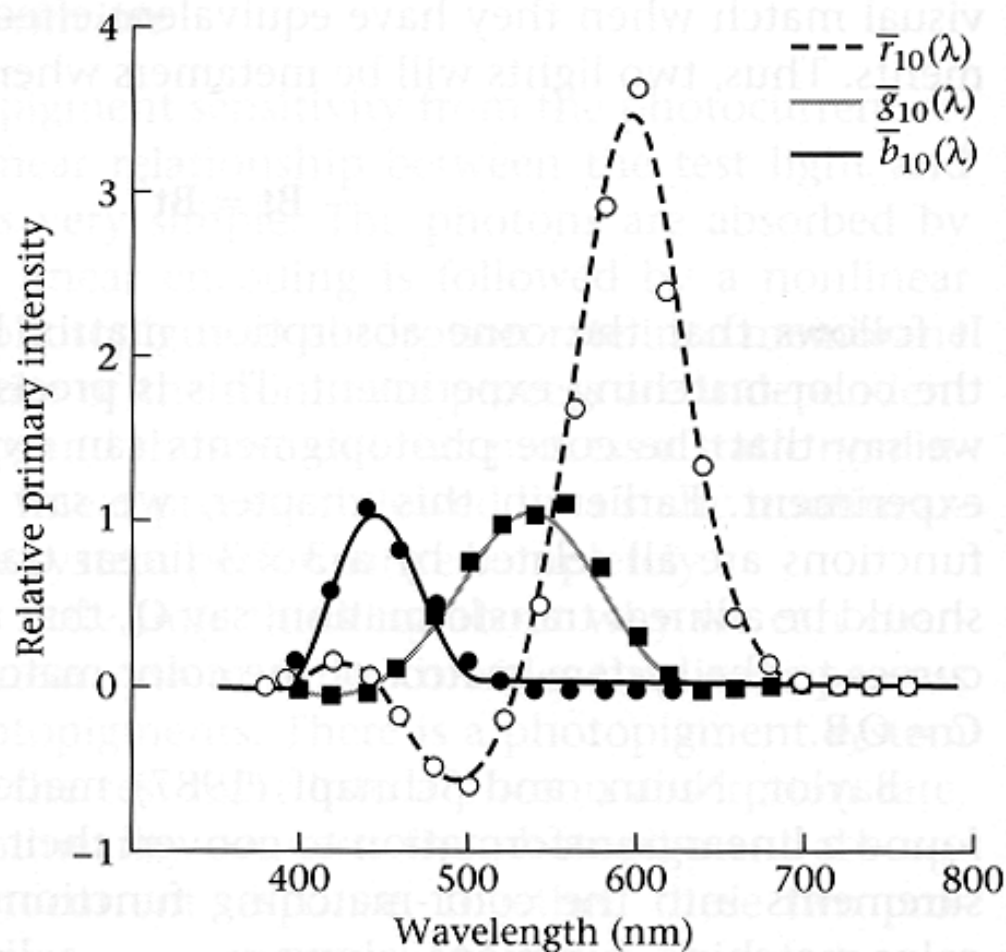
 $p_1 = 645.2 \text{ nm}$
 $p_2 = 525.3 \text{ nm}$
 $p_3 = 444.4 \text{ nm}$

RGB matching functions



Comparison of RGB matching functions with best 3x3 transformation of cone responses

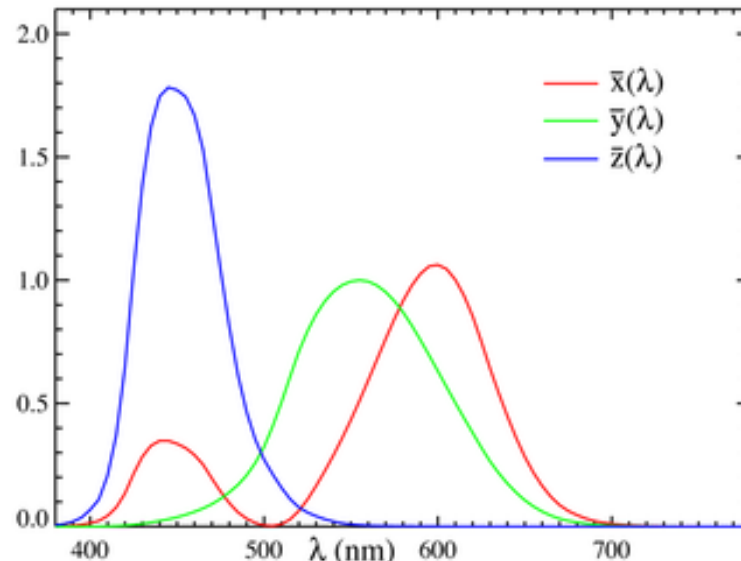
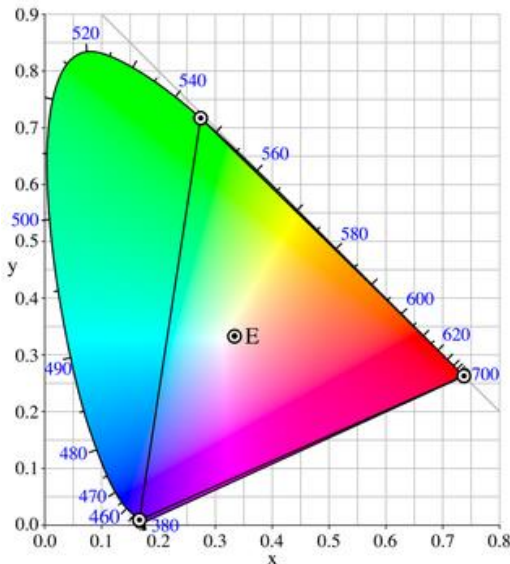
4.20 COMPARISON OF CONE PHOTOCURRENT RESPONSES AND THE COLOR-MATCHING FUNCTIONS. The cone photocurrent spectral responsivities are within a linear transformation of the color-matching functions, after a correction has been made for the optics and inert pigments in the eye. The smooth curves show the Stiles and Burch (1959) color-matching functions. The symbols show the matches predicted from the photocurrents of the three types of macaque cones. The predictions included a correction for absorption by the lens and other inert pigments in the eye. Source: Baylor, 1987.



Linear color spaces: CIE XYZ

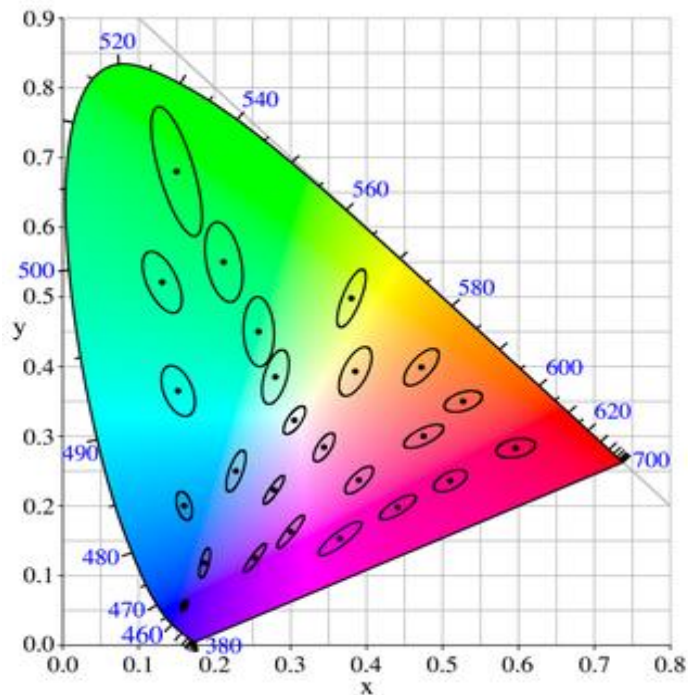
- Primaries are *imaginary*, but matching functions are everywhere positive
- The Y parameter corresponds to brightness or *luminance* of a color
- 2D visualization: draw (x,y), where $x = X/(X+Y+Z)$, $y = Y/(X+Y+Z)$

Matching functions

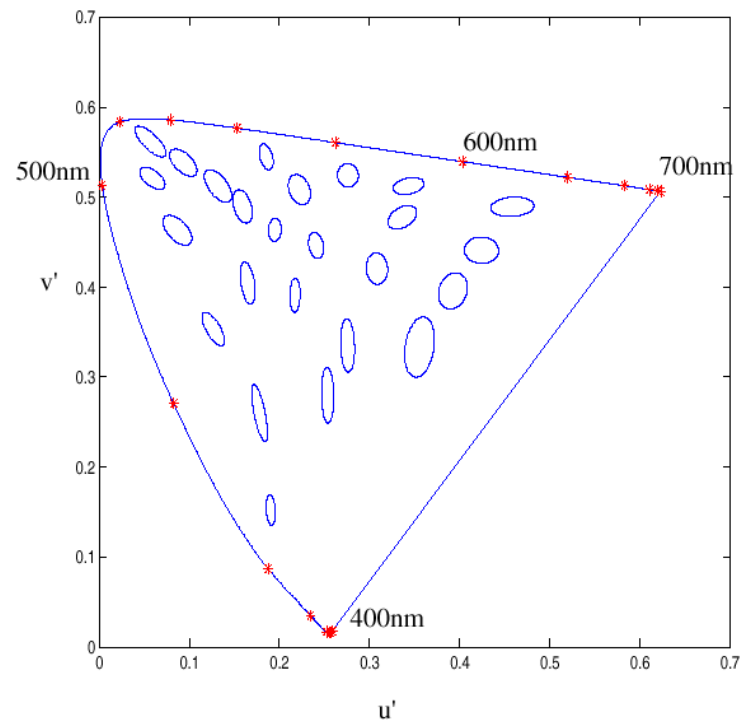


Uniform color spaces

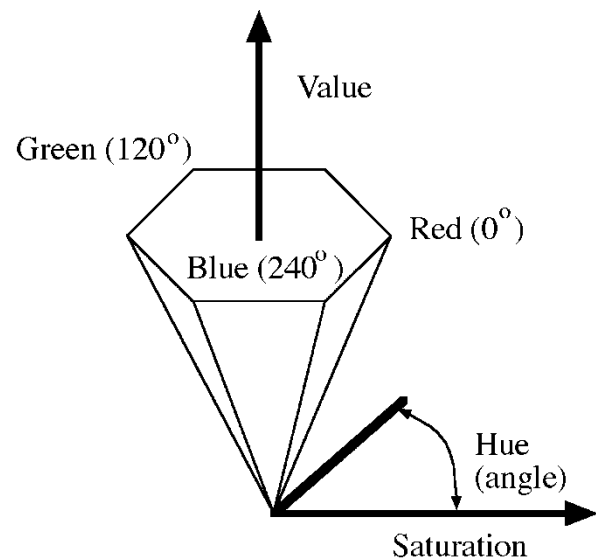
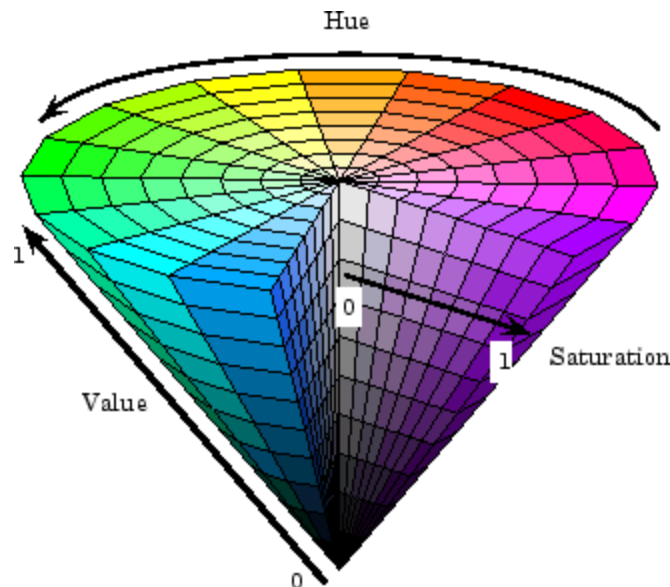
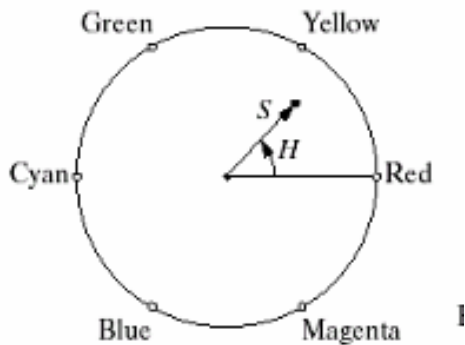
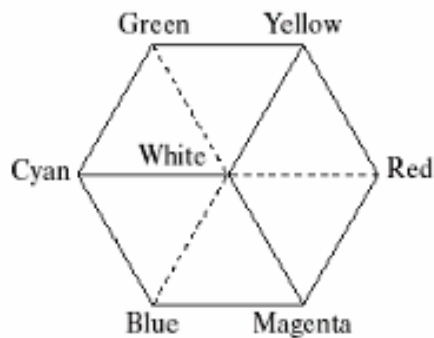
- Unfortunately, differences in x, y coordinates do not reflect perceptual color differences
- CIE $u'v'$ is a projective transform of x, y to make the ellipses more uniform



McAdam ellipses: Just noticeable differences in color



Nonlinear color spaces: HSV



- Perceptually meaningful dimensions: Hue, Saturation, Value (Intensity)
- RGB cube on its vertex

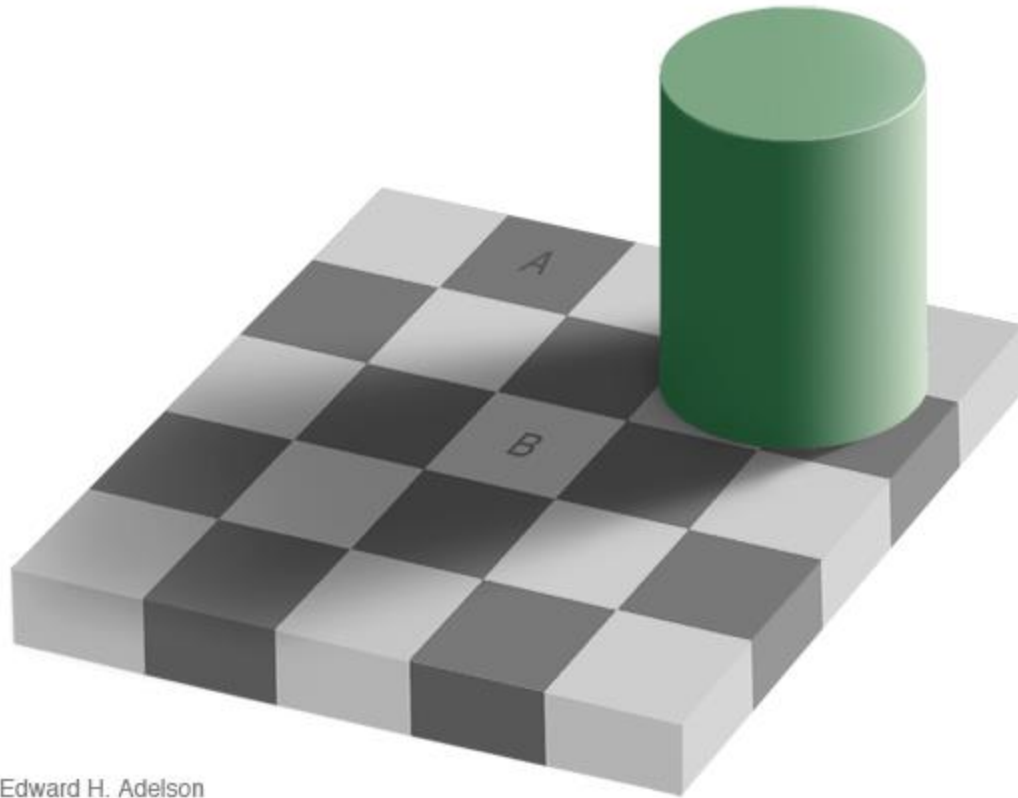
Color perception

- Color/lightness constancy
 - The ability of the human visual system to perceive the intrinsic reflectance properties of the surfaces despite changes in illumination conditions



J. S. Sargent, The Daughters of Edward D. Boit, 1882

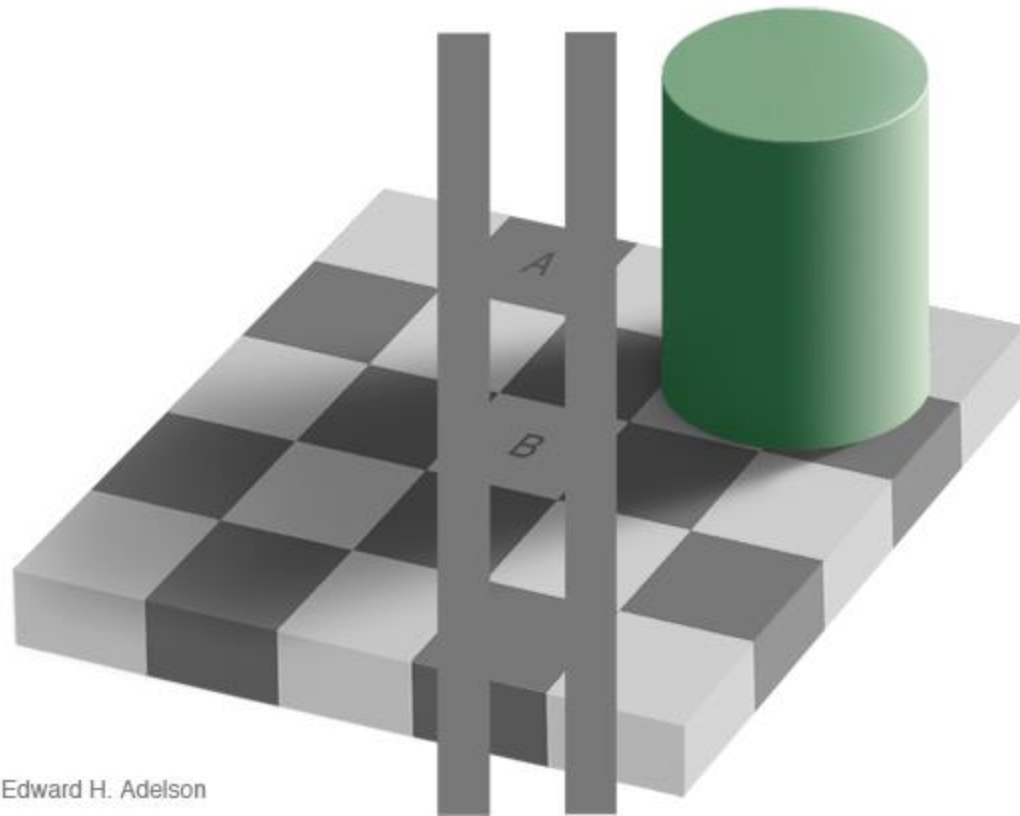
Checker shadow illusion



Edward H. Adelson

https://en.wikipedia.org/wiki/Checker_shadow_illusion

Checker shadow illusion

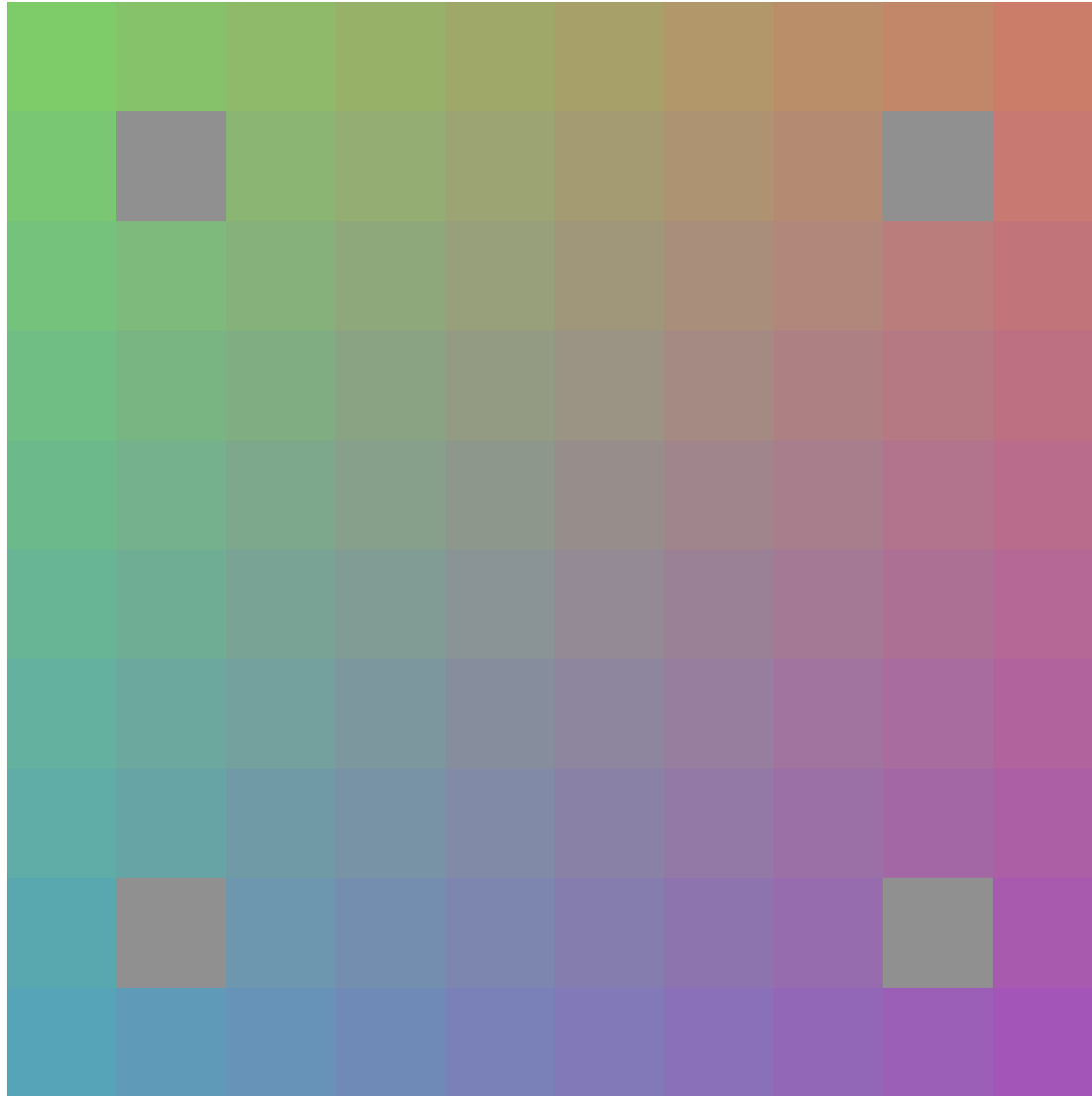


Edward H. Adelson

- Possible explanations
 - Simultaneous contrast
 - Reflectance edges vs. illumination edges

https://en.wikipedia.org/wiki/Checker_shadow_illusion

Simultaneous contrast/Mach bands



Chromatic adaptation

- The visual system changes its sensitivity depending on the luminances prevailing in the visual field
 - The exact mechanism is poorly understood
- Adapting to different brightness levels
 - Changing the size of the iris opening (i.e., the aperture) changes the amount of light that can enter the eye
 - Think of walking into a building from full sunshine
- Adapting to different color temperature
 - The receptive cells on the retina change their sensitivity
 - For example: if there is an increased amount of red light, the cells receptive to red decrease their sensitivity until the scene looks white again
 - We actually adapt better in brighter scenes: This is why candlelit scenes still look yellow

White balance

- When looking at a picture on screen or print, our eyes are adapted to the illuminant of the room, not to that of the scene in the picture
- When the white balance is not correct, the picture will have an unnatural color “cast”

incorrect white balance



correct white balance



What color is the dress?



<https://www.wired.com/2015/02/science-one-agrees-color-dress/>

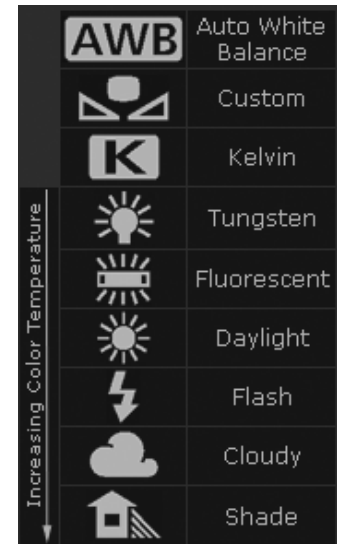
This strawberry cake has no red pixels!



<https://www.digitaltrends.com/photography/non-red-strawberries/>

White balance

- Film cameras:
 - Different types of film or different filters for different illumination conditions
- Digital cameras:
 - Automatic white balance
 - White balance settings corresponding to several common illuminants
 - Custom white balance using a reference object



White balance

- Von Kries adaptation
 - Multiply each channel by a gain factor
- Best way: gray card
 - Take a picture of a neutral object (white or gray)
 - Deduce the weight of each channel
 - If the object is recoded as r_w , g_w , b_w
use weights $1/r_w$, $1/g_w$, $1/b_w$



White balance

- Without gray cards: we need to “guess” which pixels correspond to white objects
- Gray world assumption
 - The image average r_{ave} , g_{ave} , b_{ave} is gray
 - Use weights $1/r_{ave}$, $1/g_{ave}$, $1/b_{ave}$
- Brightest pixel assumption
 - Highlights usually have the color of the light source
 - Use weights inversely proportional to the values of the brightest pixels
- Gamut mapping
 - Gamut: convex hull of all pixel colors in an image
 - Find the transformation that matches the gamut of the image to the gamut of a “typical” image under white light
- Use image statistics, learning techniques

Mixed illumination

- When there are several types of illuminants in the scene, different reference points will yield different results



Reference: moon



Reference: stone

Spatially varying white balance



Input



Alpha map



Output

E. Hsu, T. Mertens, S. Paris, S. Avidan, and F. Durand, [Light Mixture Estimation for Spatially Varying White Balance](#), SIGGRAPH 2008

Color cues for image understanding

Color-based image matching

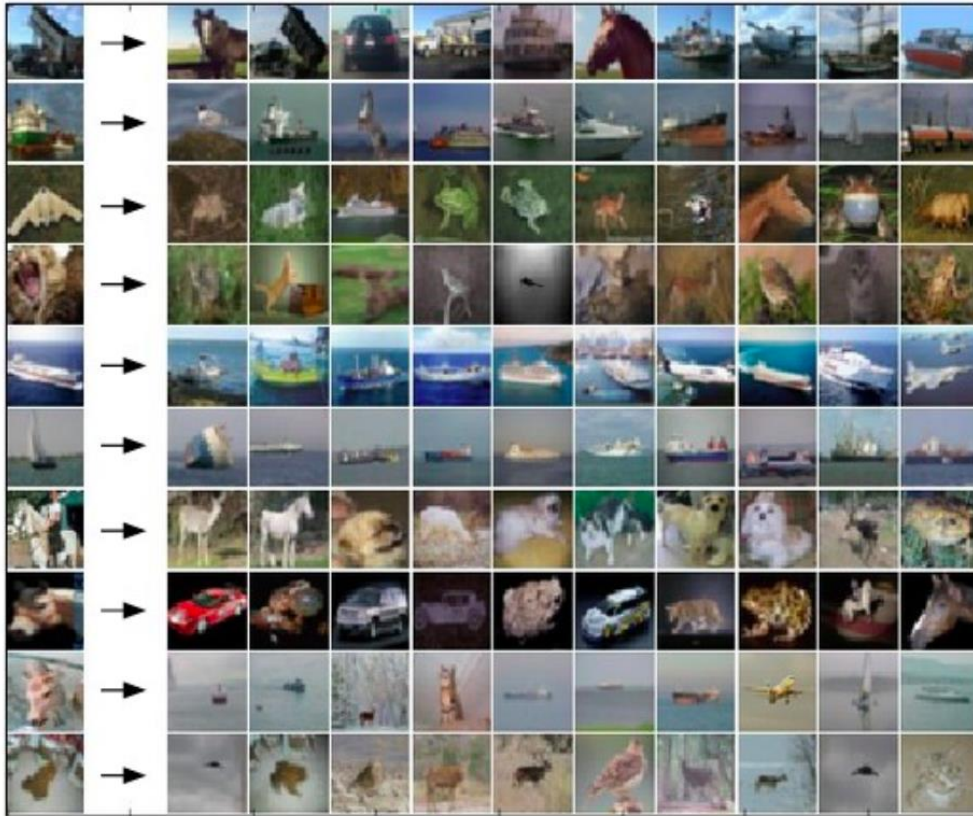


Image segmentation